

Image and Video Generations

Lecture 6: DDIM Inversion / Score Distillation

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DDIM Inversion

Mokady, Ron, et al., Null-text Inversion for Editing Real Images using Guided Diffusion Models, CVPR 2023.

DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \mathbf{Q}$. Write $\mathbf{x}_{0|t}$ in terms of $\mathbf{x}_t$ and $\hat{\mathbf{e}}_{\theta}(\mathbf{x}_t, t)$.
2. $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\hat{\mathbf{e}}_{\theta}(\mathbf{x}_t, t)$.
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t$.

DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$
2. $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t).$
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$
4. $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t.$

DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$

2. $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t).$

3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$

4. $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t.$

DDIM 的一般形式，所以有 σ_t
 $\sigma_t = 0$ 就是我們熟知的 deterministic DDIM

DDIM Deterministic Sampling

If $\sigma_t = 0$ for all t ,

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right)$
2. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t).$

Q. What is $\mathbf{x}_{t-1}$ with respect to $\mathbf{x}_t$ and $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$?

DDIM Deterministic Sampling

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left[\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \right] + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

預測的 $\mathbf{x}_{0|t}$ 指向 $\mathbf{x}_t$ 的方向

DDIM Inversion

- For **deterministic** sampling, the mapping from x_T to x_0 is **fixed**.
- Then, how can we also compute the **inverse mapping** from x_0 to x_T ?

給定一張照片 (x_0)，找到一個 noise (x_T)
經過 deterministic DDIM 反向去噪最終會得到原本的 x_0



DDIM Inversion – 思路一

交換 $\mathbf{x}_{t-1}$ 與 $\mathbf{x}_t$ 的位置，用 $\mathbf{x}_{t-1}$ 去計算 $\mathbf{x}_t$ ：

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

$$\Rightarrow \mathbf{x}_t = \sqrt{\frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \mathbf{x}_{t-1} + \left(\sqrt{\frac{\bar{\alpha}_{t-1}(1 - \bar{\alpha}_t)}{\bar{\alpha}_{t-1}}} - \sqrt{\frac{\bar{\alpha}_t(1 - \bar{\alpha}_{t-1})}{\bar{\alpha}_{t-1}}} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

$$\mathbf{x}_t = \sqrt{\frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \mathbf{x}_{t-1} + \sqrt{\bar{\alpha}_t} \left(\sqrt{\frac{1}{\bar{\alpha}_t}} - 1 - \sqrt{\frac{1}{\bar{\alpha}_{t-1}}} - 1 \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

但是目前我們還不知道 $\mathbf{x}_t$

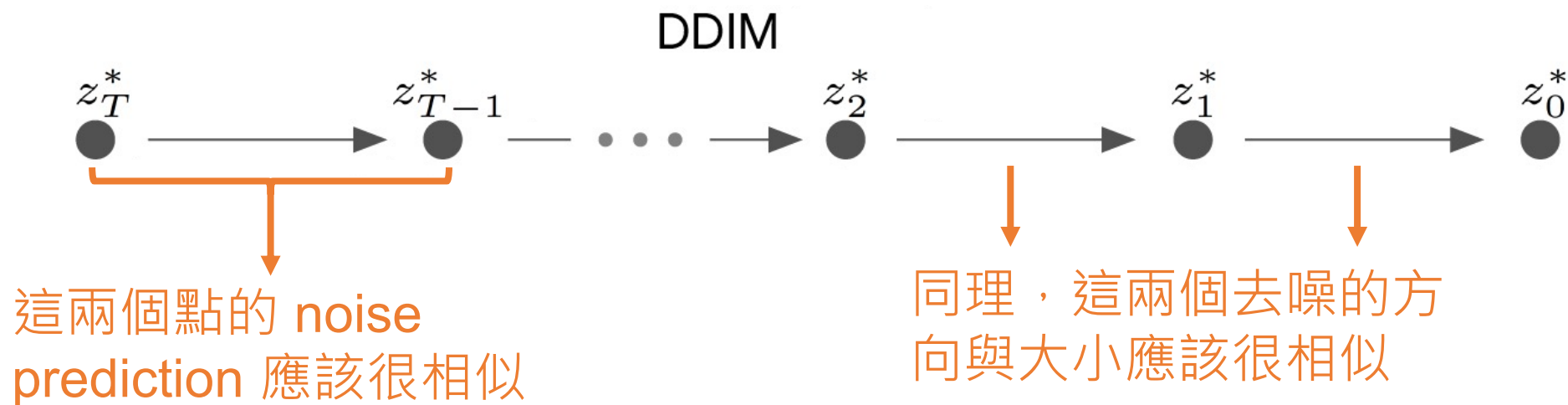
只能估計 $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$ ，例如 $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$ 或 $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

Q. 為什麼可以這樣估計？

A. 線性假設



Input Image

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

DDIM Inversion

z_0^*
●



Input Image

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

DDIM Inversion

我們想代入這個式子來得到 $\mathbf{x}_1$

$$\mathbf{x}_t = \sqrt{\frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \mathbf{x}_{t-1} + \sqrt{\bar{\alpha}_t} \left(\sqrt{\frac{1}{\bar{\alpha}_t} - 1} - \sqrt{\frac{1}{\bar{\alpha}_{t-1}} - 1} \right) \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

z_0^*
●



Input Image

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

DDIM Inversion

但我們沒有 $\hat{\epsilon}_{\theta}(\mathbf{x}_1, 1)$

z_0^*
●



Input Image

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

DDIM Inversion

但我們沒有 $\hat{\epsilon}_{\theta}(\mathbf{x}_1, 1)$
用 $\hat{\epsilon}_{\theta}(\mathbf{x}_0, 0)$ 或 $\hat{\epsilon}_{\theta}(\mathbf{x}_0, 1)$ 去近似

z_0^*
●



Input Image

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

DDIM Inversion



代進式子走出加噪一步：

$$\mathbf{x}_t = \sqrt{\frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \mathbf{x}_{t-1} + \sqrt{\bar{\alpha}_t} \left(\sqrt{\frac{1}{\bar{\alpha}_t} - 1} - \sqrt{\frac{1}{\bar{\alpha}_{t-1}} - 1} \right) \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1)$$



Input Image

DDIM Inversion – 思路一

$$\hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t-1) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \text{ 或 } \hat{\epsilon}_{\theta}(\mathbf{x}_{t-1}, t) \approx \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



Input Image

DDIM Deterministic Sampling

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

$$= \sqrt{\bar{\alpha}_{t-1}} \left[\sqrt{\frac{1}{\bar{\alpha}_t}} \mathbf{x}_t + \left(\sqrt{\frac{1}{\bar{\alpha}_{t-1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right]$$

DDIM Inversion – 思路二

Key idea: For the forward process with small time intervals, approximate $(\mathbf{x}_{t+1} - \mathbf{x}_t)$ by simply replacing $(t - 1)$ with $(t + 1)$ in the $(\mathbf{x}_{t-1} - \mathbf{x}_t)$ formulation:

$$\mathbf{x}_{t-1} - \mathbf{x}_t = \sqrt{\bar{\alpha}_{t-1}} \left[\left(\sqrt{\frac{1}{\bar{\alpha}_t}} - \sqrt{\frac{1}{\bar{\alpha}_{t-1}}} \right) \mathbf{x}_t + \left(\sqrt{\frac{1}{\bar{\alpha}_{t-1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right]$$

DDIM Inversion – 思路二

Key idea: For the forward process with small time intervals, approximate $(\mathbf{x}_{t+1} - \mathbf{x}_t)$ by simple replacing $(t - 1)$ with $(t + 1)$ in the $(\mathbf{x}_{t-1} - \mathbf{x}_t)$ formulation:

$$\mathbf{x}_{t+1} - \mathbf{x}_t = \sqrt{\bar{\alpha}_{t+1}} \left[\left(\sqrt{\frac{1}{\bar{\alpha}_t}} - \sqrt{\frac{1}{\bar{\alpha}_{t+1}}} \right) \mathbf{x}_t + \left(\sqrt{\frac{1}{\bar{\alpha}_{t+1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right]$$

DDIM Inversion – 思路二

Q. 為什麼可以直接把 $t - 1$ 替換成 $t + 1$?

DDIM Inversion – 思路二

Q. 為什麼可以直接把 $t - 1$ 替換成 $t + 1$?

A. 線性假設



Input Image

DDIM Inversion

第一種思路

$$\mathbf{x}_t = \sqrt{\frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \mathbf{x}_{t-1} + \sqrt{\bar{\alpha}_t} \left(\sqrt{\frac{1}{\bar{\alpha}_t} - 1} - \sqrt{\frac{1}{\bar{\alpha}_{t-1}} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_{t-1}, t-1)$$

第二種思路

$$\mathbf{x}_{t+1} - \mathbf{x}_t = \sqrt{\bar{\alpha}_{t+1}} \left[\left(\sqrt{\frac{1}{\bar{\alpha}_t} - 1} - \sqrt{\frac{1}{\bar{\alpha}_{t+1}} - 1} \right) \mathbf{x}_t + \left(\sqrt{\frac{1}{\bar{\alpha}_{t+1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right]$$

變數變換 $t \leftarrow t + 1$

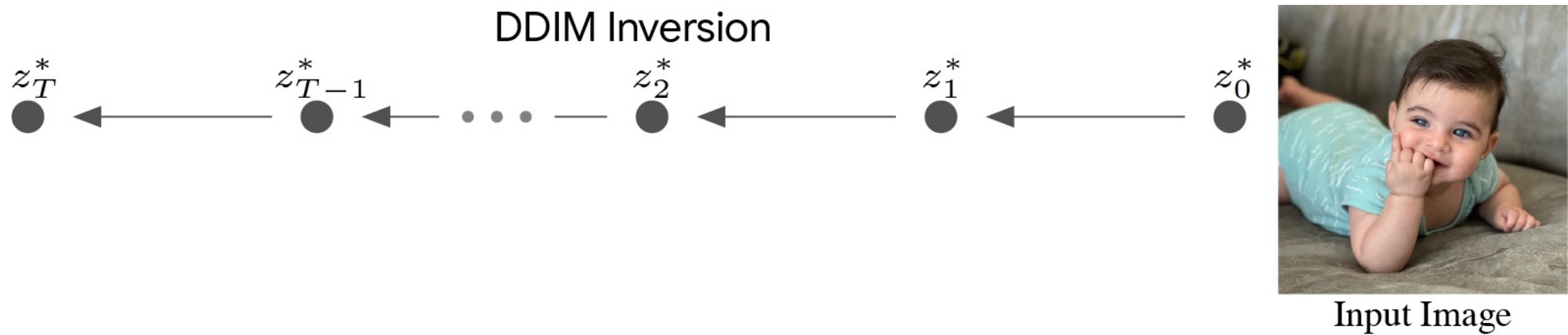
$$\Rightarrow \mathbf{x}_{t+1} = \sqrt{\frac{\bar{\alpha}_{t+1}}{\bar{\alpha}_t}} \mathbf{x}_t + \sqrt{\bar{\alpha}_{t+1}} \left(\sqrt{\frac{1}{\bar{\alpha}_{t+1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

$$\Rightarrow \mathbf{x}_{t+1} = \sqrt{\frac{\bar{\alpha}_{t+1}}{\bar{\alpha}_t}} \mathbf{x}_t + \sqrt{\bar{\alpha}_{t+1}} \left(\sqrt{\frac{1}{\bar{\alpha}_{t+1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$$

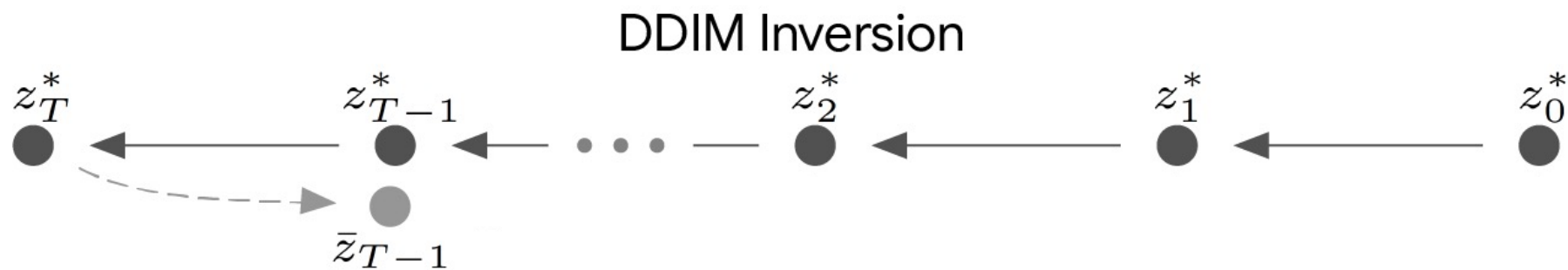
兩種思路其實完全相同

因為他們都基於相同的核心假設：**noise prediction 的連續性**

DDIM Inversion

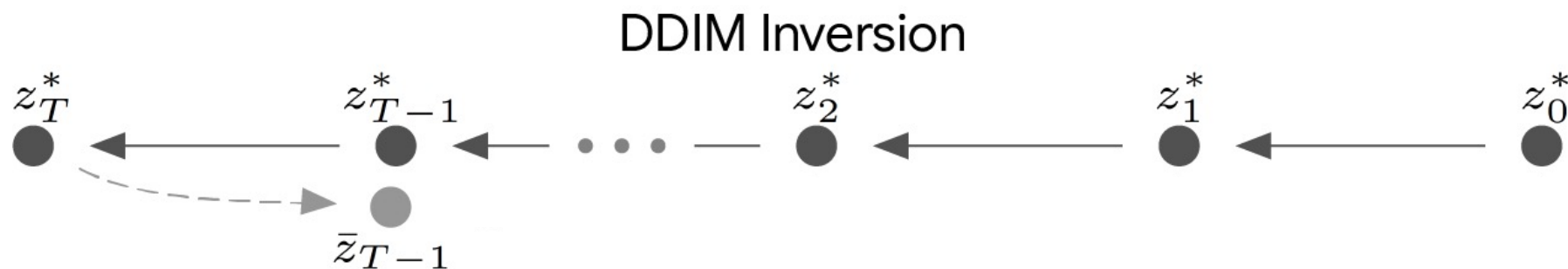


DDIM Inversion



Input Image

DDIM Inversion

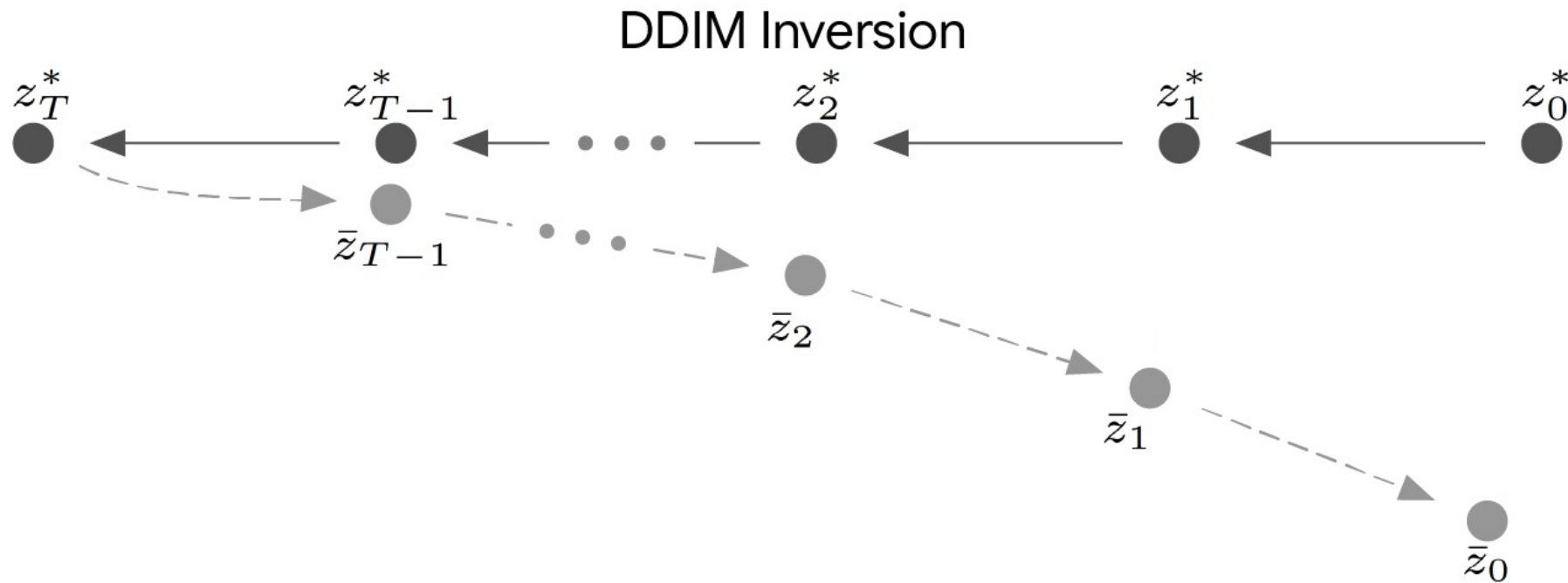


Input Image

Q. 為什麼這邊會有誤差？

A. Noise prediction 的相似性只是一個**假設**
(尤其是 CFG 很大的時候)

DDIM Inversion



Input Image



Initial Inversion

Q. 為什麼 DDIM inversed noise 的重建結果會跟輸入影像差這麼多？

A. 因為前述的假設會引入誤差（尤其是 CFG 很大的時候）
這些誤差會在每一次的 denoising step 中逐步累積放大

DDIM Inversion

重建的品質也與總 inverse time steps 有關

每個 x_t 之間的時間隔越小（總 inverse **time steps** 數越大），Noise prediction 的相似性假設越準確



DDIM Inversion

Q. 所以間隔越小 / 總 inversion time steps 數越高越好嗎？

A. Not always

Trade-off between 每步誤差 and 累積誤差

DDIM Inversion

Result of inversion ($x_0 \rightarrow x_T$) and generation ($x_T \rightarrow x_0$)



Input image



Regenerated image

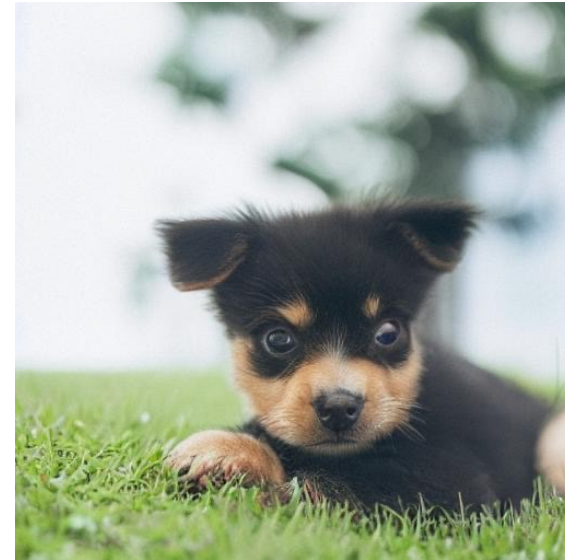
Time steps = 200 , 也就是每個 x_t 之間的時間隔較小

DDIM Inversion

Inversion fails when the number of time steps is too small (when the time intervals are too large).



Input image



Regenerated image
when the number of time steps is 50.

Image Editing Using DDIM Inversion

1. Perform DDIM inversion using the **original prompt** in CFG.
2. Perform reverse processing using a **new prompt** in CFG.



"A photograph of a puppy"



"A photograph of a grey cat"

Image Editing Using DDIM Inversion



“A photograph of a face”



“A photograph of a face with sunglasses”



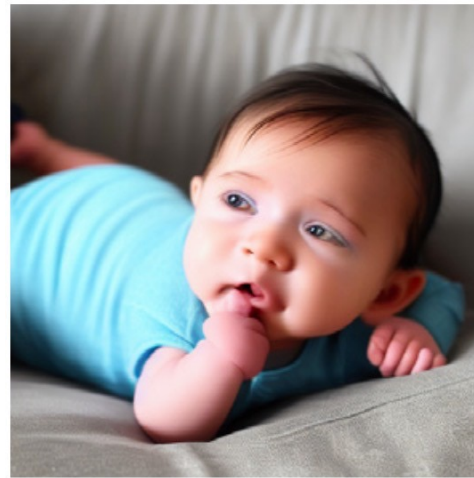
“Acrylic palette knife painting of a face, colorful”

Image Editing Using DDIM Inversion

- Good image editing can be achieved by using a **high CFG weight** for both inversion and generation.
- However, inversion tends to fail when CFG weight ω is high.



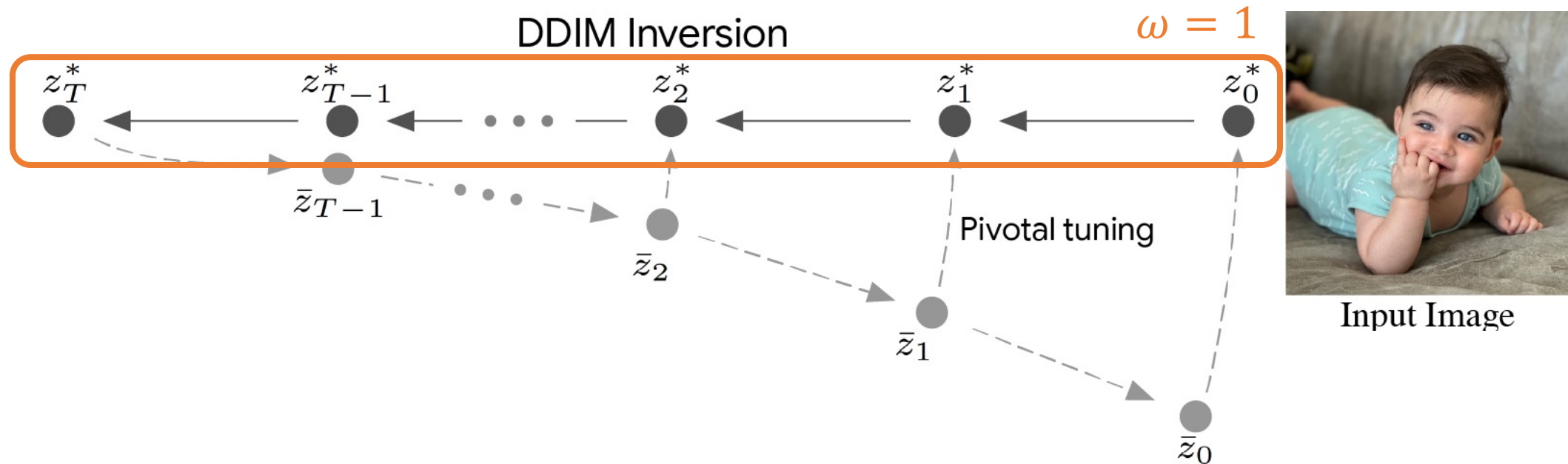
Input image



Regenerated image with $\omega = 7.5$.

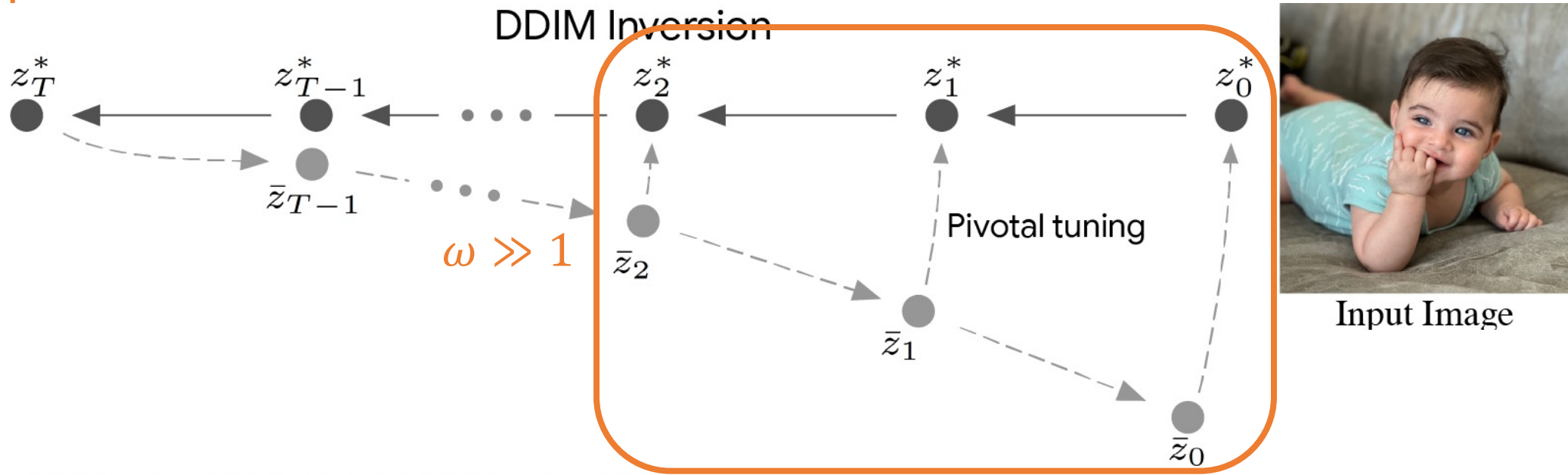
Null-Text Inversion

- Perform inversion with $\omega = 1$. Let the latent variables $\{x_t^*\}_{t=1,\dots,T}$ be **pivots**.



Null-Text Inversion

- Perform inversion with $\omega = 1$. Let the latent variables $\{x_t^*\}_{t=1,\dots,T}$ be **pivots**.
- Perform inversion with $\omega \gg 1$ while enforcing the latent variable x_t to be close to the corresponding latent variable x_t^* **while tuning some parameters**.

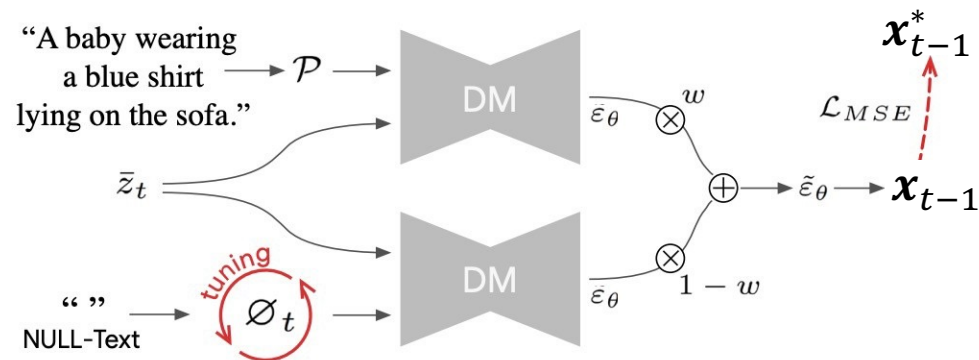


Null-Text Inversion

Which parameter to be tuned? **Null text prompt ϕ !**

When $\mathbf{x}_{t-1}(\mathbf{x}_t, c, \phi)$ indicating the computation of $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$, input prompt c , and null prompt ϕ , minimize

$$\|\mathbf{x}_{t-1}^* - \mathbf{x}_{t-1}(\mathbf{x}_t, c, \phi)\|_2^2$$

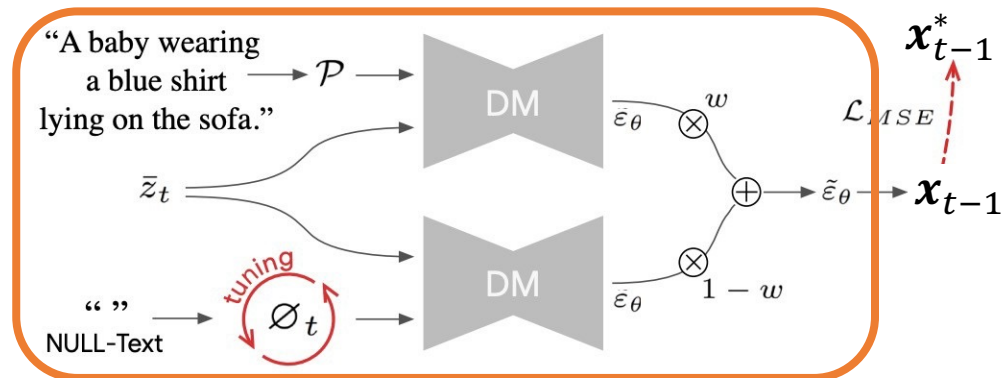


Null-Text Inversion

Which parameter to be tuned? **Null text prompt ϕ !**

When $\mathbf{x}_{t-1}(\mathbf{x}_t, c, \phi)$ indicating the computation of $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$, input prompt c , and null prompt ϕ , minimize

$$\|\mathbf{x}_{t-1}^* - \mathbf{x}_{t-1}(\mathbf{x}_t, c, \phi)\|_2^2$$



CFG

Null-Text Inversion



Input image



Regenerated image
with $\omega = 7.5$.



Regenerated image
with $\omega = 7.5$
and null-text inversion

Null-Text Inversion

Comparison of inversion methods



Null-Text Inversion

Comparison of inversion methods



但要針對每張測試影像 fine-tune 過

Null-Text Inversion

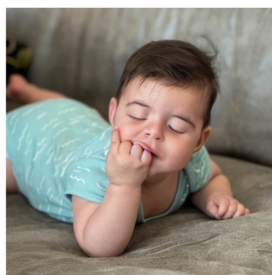
Input caption: “A baby wearing a blue shirt lying on the sofa.”



Input Image



“...blond baby...”



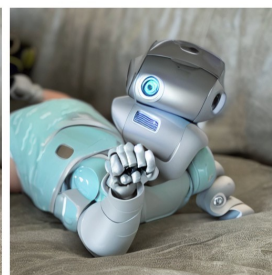
“...sleeping baby...”



“...golden shirt...”



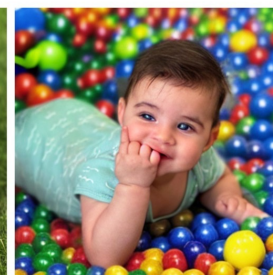
“... floral shirt...”



baby → robot



sofa → grass



sofa → ball pit

Input caption: “A man in glasses eating a doughnut in the park.”



Input Image



“...red haired man...”



glasses → sunglasses



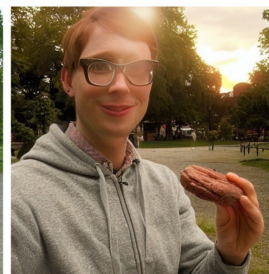
“...angry man...”



doughnut → pizza



glasses → Joker mask



“...the park at sunset.”



park → desert

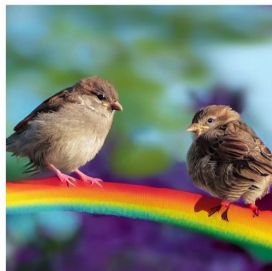
但要針對每張測試影像 fine-tune 過

Null-Text Inversion

Input caption: “Two birds sitting on a branch.”



Input Image



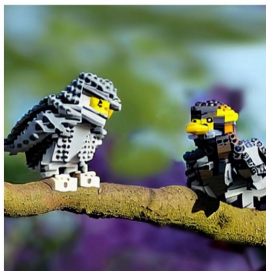
branch → rainbow



branch → metal pole



branch → electric cable



“...Lego birds”



“...crochet birds”



“...origami birds”



“...jello birds”

Input caption: “Cake on a table.”



Input Image



“Chocolate cake...”



“Strawberry cake...”



“Spinach moss cake...”



“Macaroni cake...”



“Purple neon cake...”



“Monster cake...”

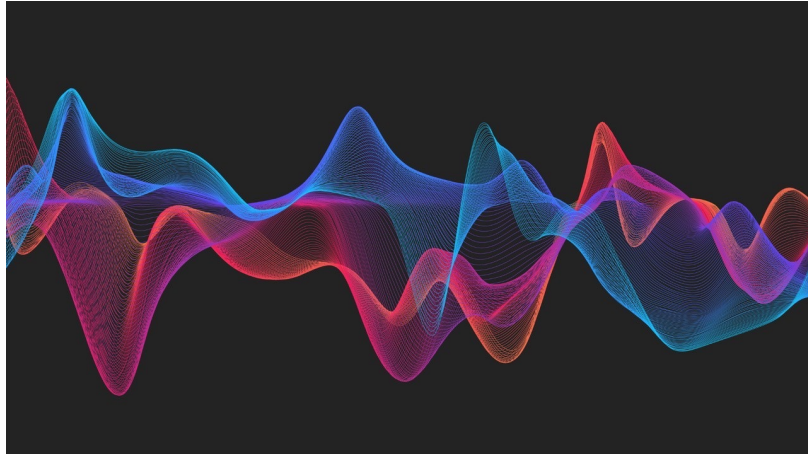


“Pepperoni cake...”

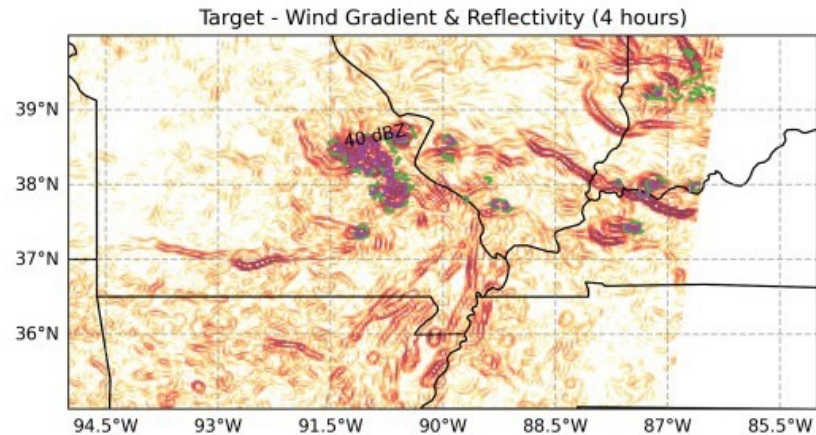
但要針對每張測試影像 fine-tune 過

Score Distillation

Diffusion Models for Various Modalities



<https://aibusiness.com/nlp/create-unique-sounds-with-meta-s-new-audio-box-ai>



https://research.nvidia.com/publication/2024-08_kilometer-scale-convection-allowing-model-emulation-using-generative-diffusion

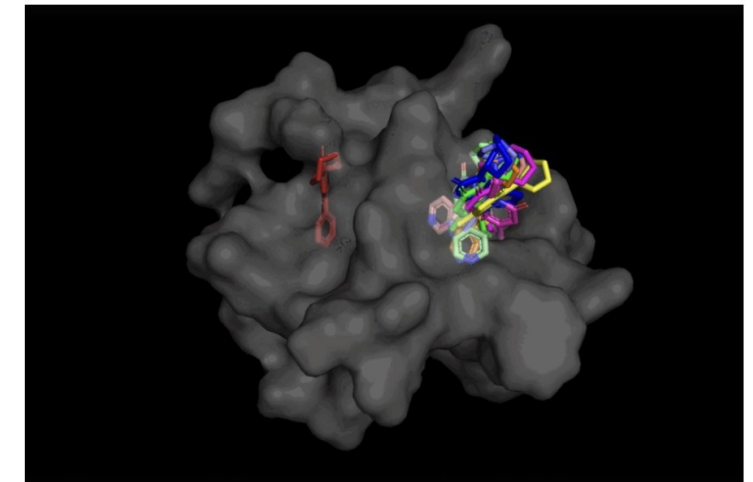


Alexanderson et al., Listen, Denoise, Action! Audio-Driven Motion Synthesis with Diffusion Models, SIGGRAPH 2023.

Speeding up drug discovery with diffusion generative models

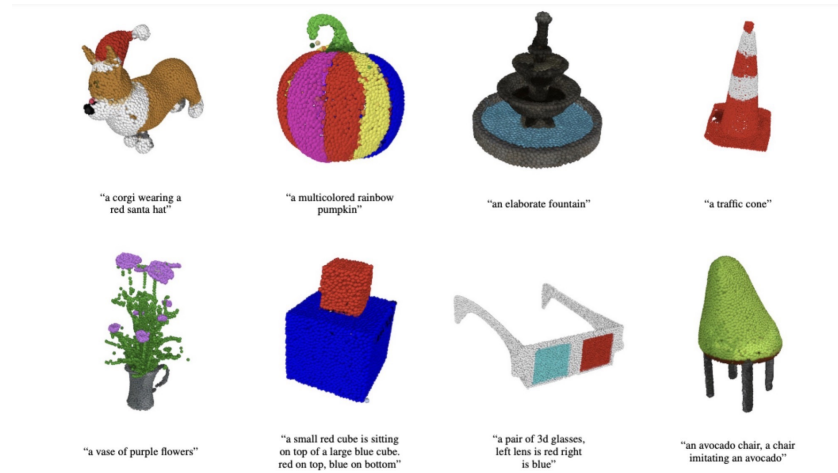
MIT researchers built DiffDock, a model that may one day be able to find new drugs faster than traditional methods and reduce the potential for adverse side effects.

Alex Ouyang | Abdul Latif Jameel Clinic for Machine Learning in Health
March 31, 2023

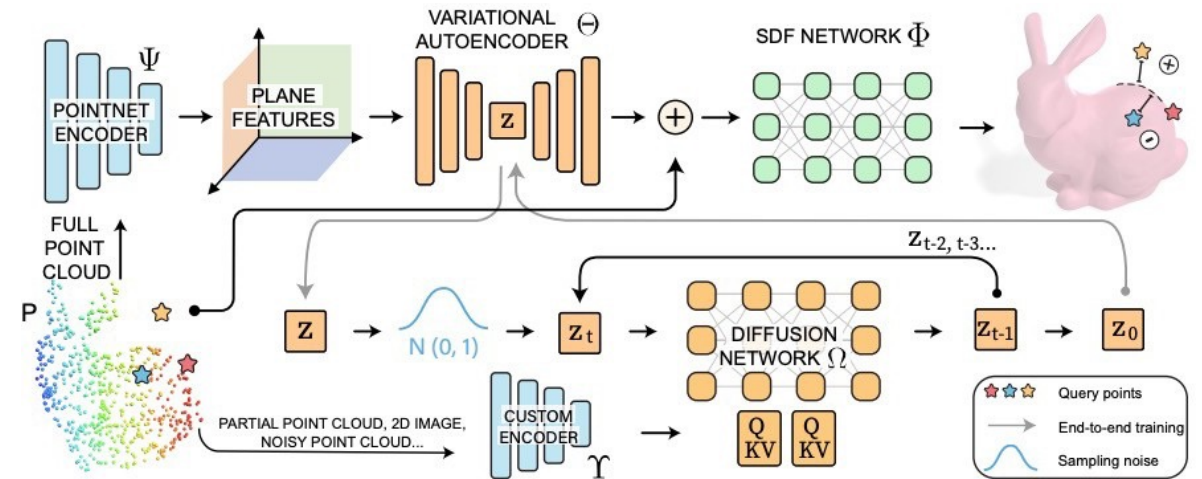


<https://news.mit.edu/2023/speeding-drug-discovery-with-diffusion-generative-models-diffdock-0331>

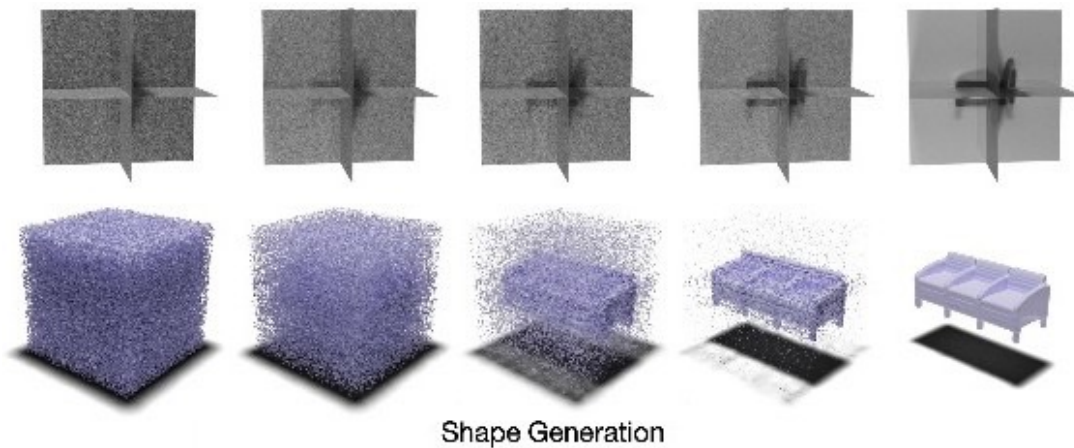
Diffusion Models for 3D



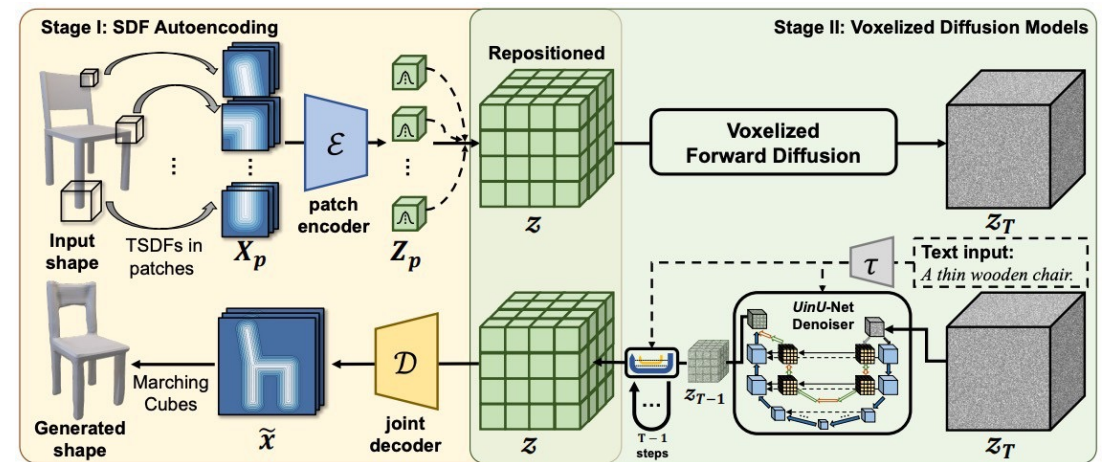
Point-E, Nichol et al., 2022.



DiffusionSDF, Chou et al., 2023

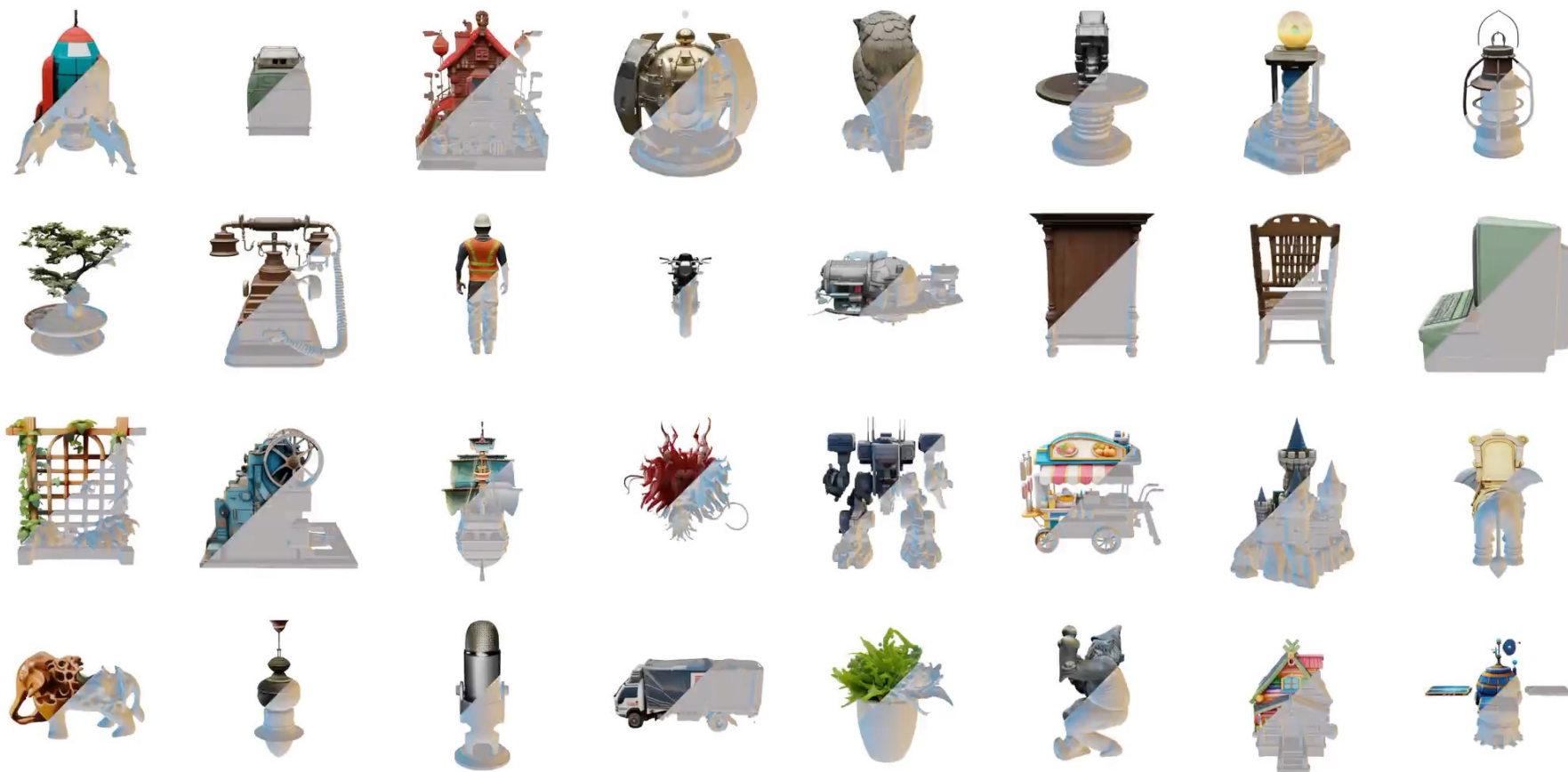


NFD, Shue et al., 2022.

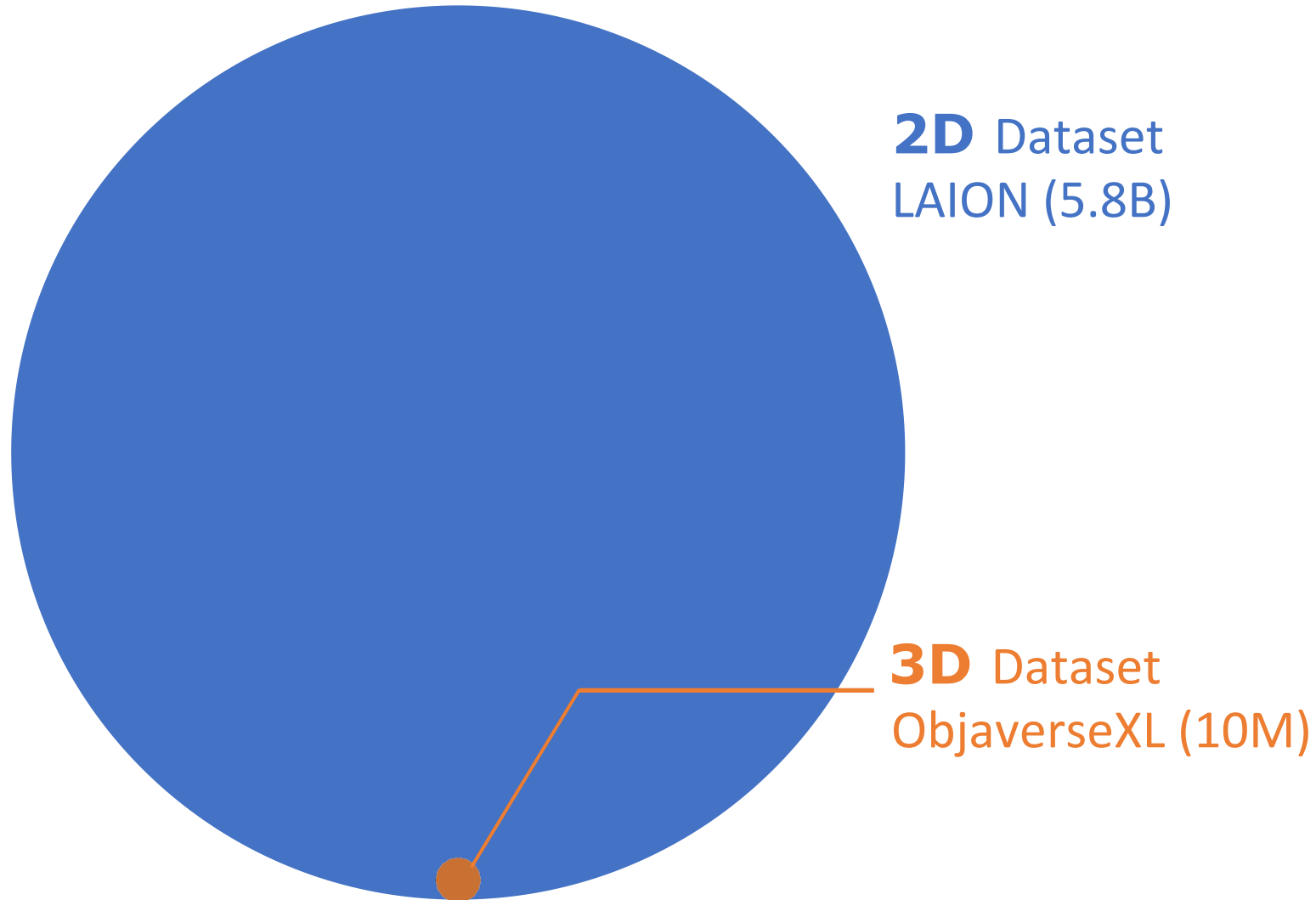


Diffusion-SDF, Li et al., 2022.

Diffusion Models for 3D



Challenge: Data Deficiency



Imaginable 3D Shapes



"frog wearing a sweater"



"eggshell broken in two
with an adorable chick
standing next to it"



"ghost eating a hamburger"



"a pig wearing a backpack"

Knowledge Distillation

Can we leverage a pretrained image diffusion model to generate other visual content?



3D

DreamFusion (Poole et al.)



Vector Image

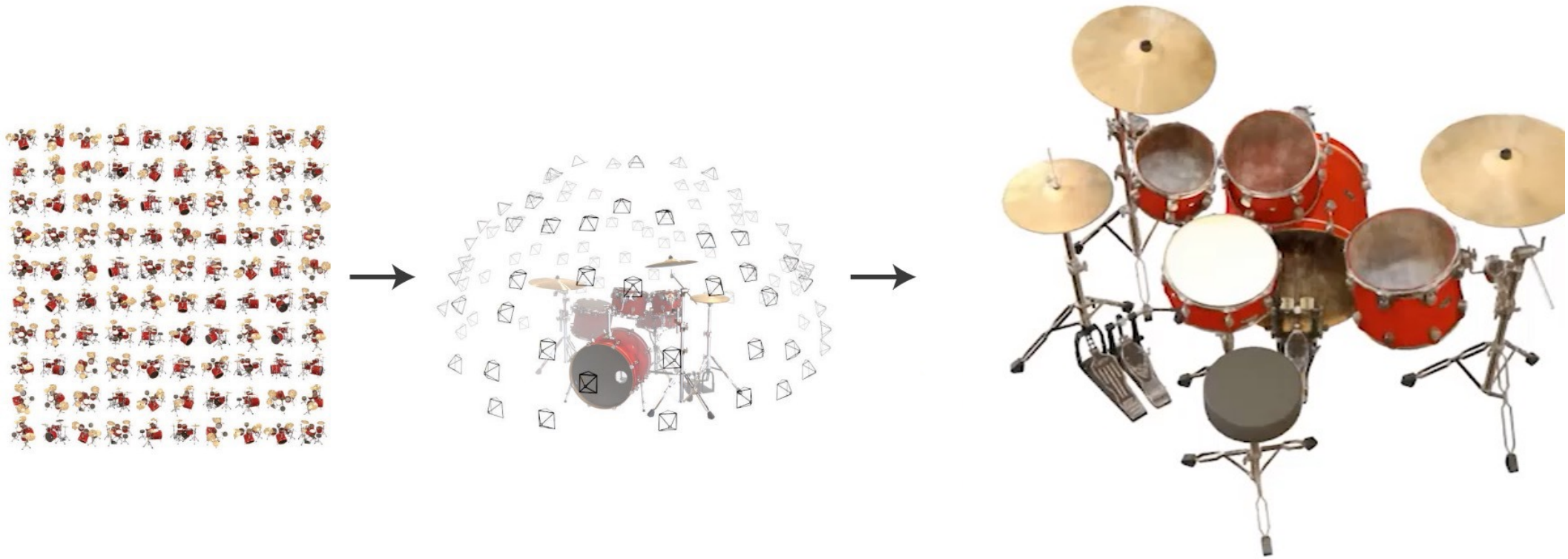
VectorFusion (Jain et al.)



Mesh Texture

Paint-it (Kim et al.)

Off-Topic: Multi-View 3D Reconstruction



NeRF Results



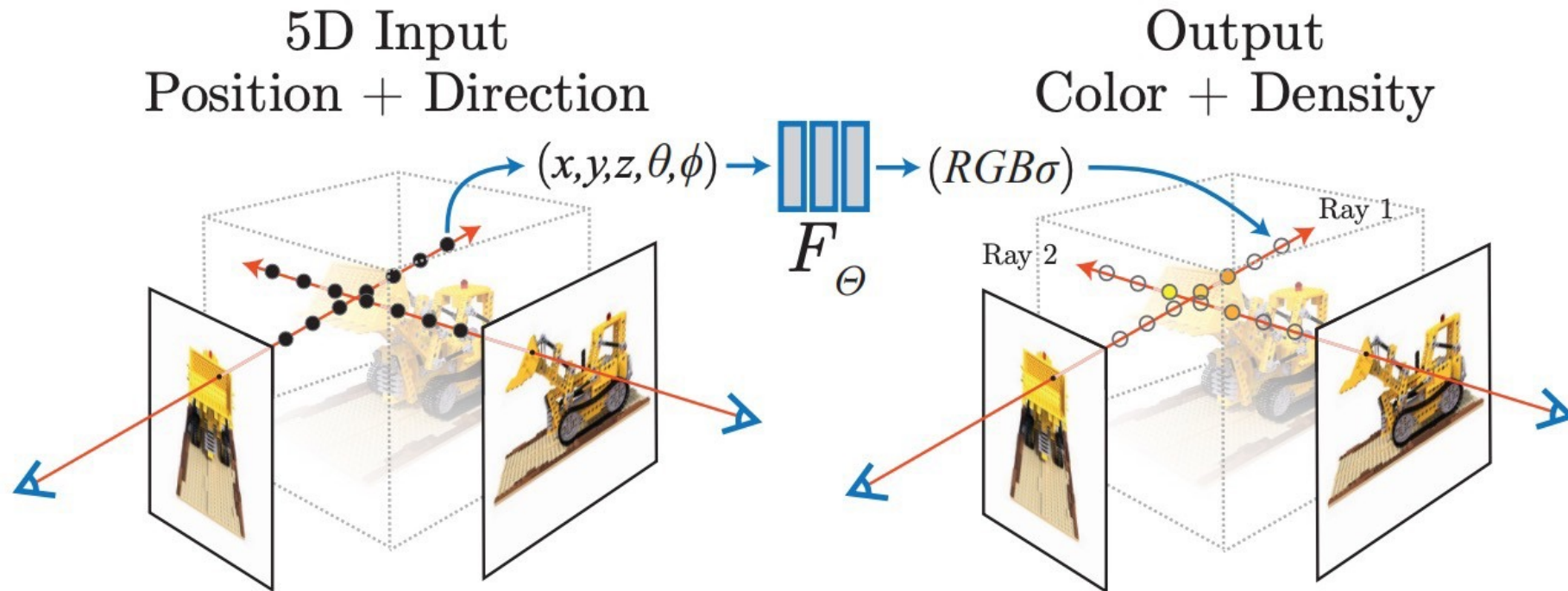
NeRF Results





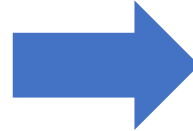
3D Representations: NeRF

Neural Radiance Fields (NeRF): Implicit representation



3D Representations: Gaussian Splats

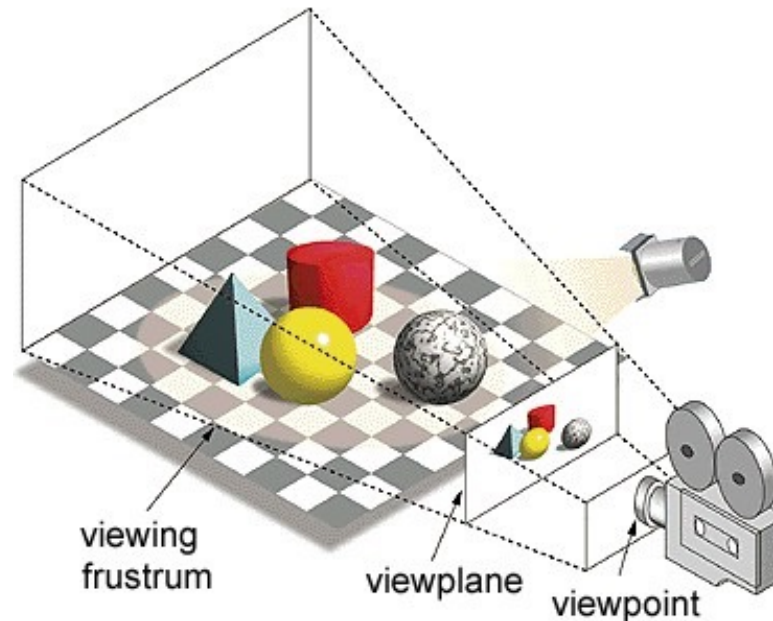
Gaussian Splats (GS):
Explicit representation



Rendering

In computer graphics, rendering is a function $g(\theta; c)$ generating an image from a 3D representation θ and a specific viewpoint c .

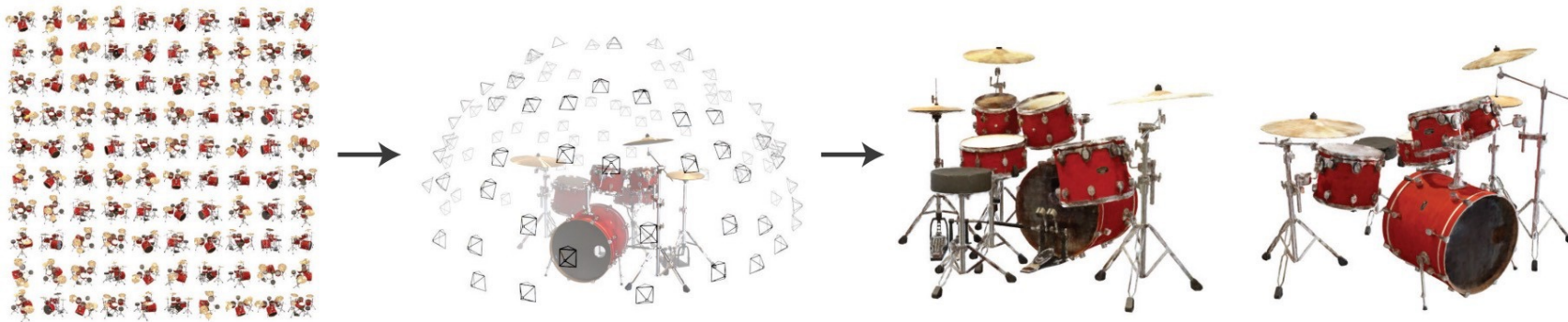
From Computer Desktop Encyclopedia
Reprinted with permission.
© 1998 Intergraph Computer Systems



3D Reconstruction via Optimization

- Let $\{\mathbf{x}_i\}_{i=1,\dots,N}$ be given images from viewpoints $\{c_i\}_{i=1,\dots,N}$.
- For each viewpoint, perform gradient descent on the 3D representation $\boldsymbol{\theta}$ using the following gradient:

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{NeRF}}(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \|g(\boldsymbol{\theta}; c_i) - \mathbf{x}_i\|^2$$



Challenges in 3D Reconstruction

- How many images are needed for accurate 3D reconstruction?
- Is it possible to achieve 3D reconstruction using only a few images?

Few-Shot NeRF

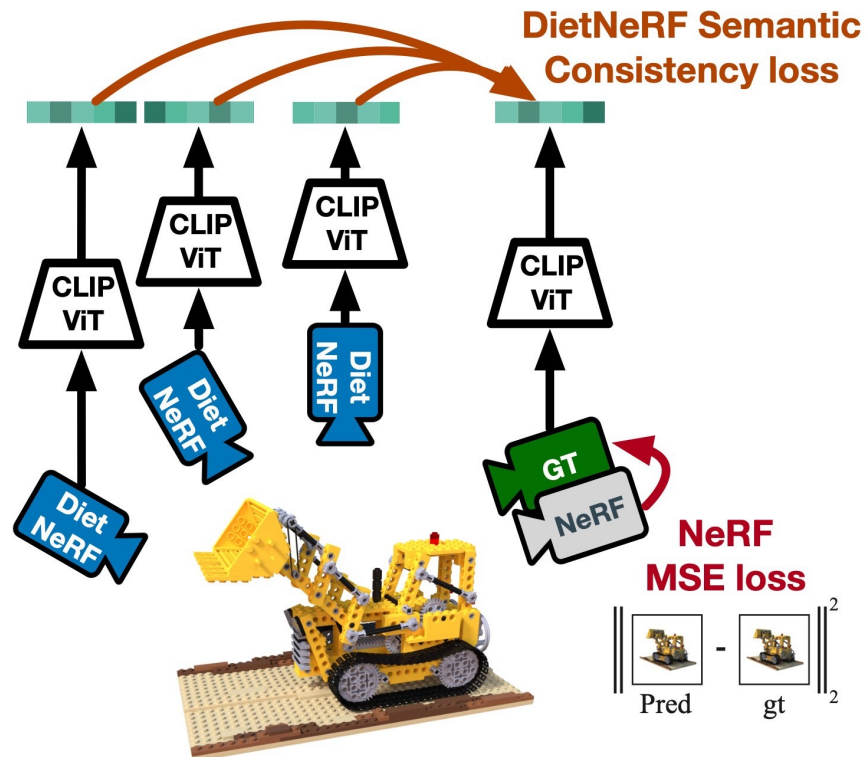
FrugalNeRF



Few-Shot NeRF

DietNeRF

Use **priors** learned by a **text-image model**: CLIP



*“a bulldozer is a bulldozer
from any perspective”*

CLIP

"A beautiful
painting of a
dog riding a
dolphin"

+

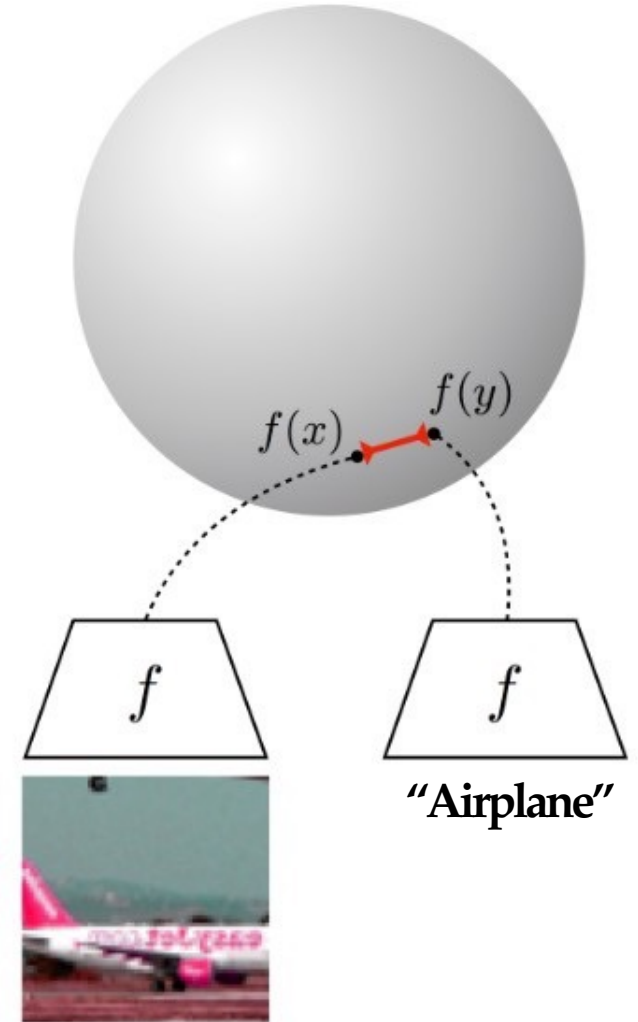


= CLIP
70.7%

STEP 500

CLIP

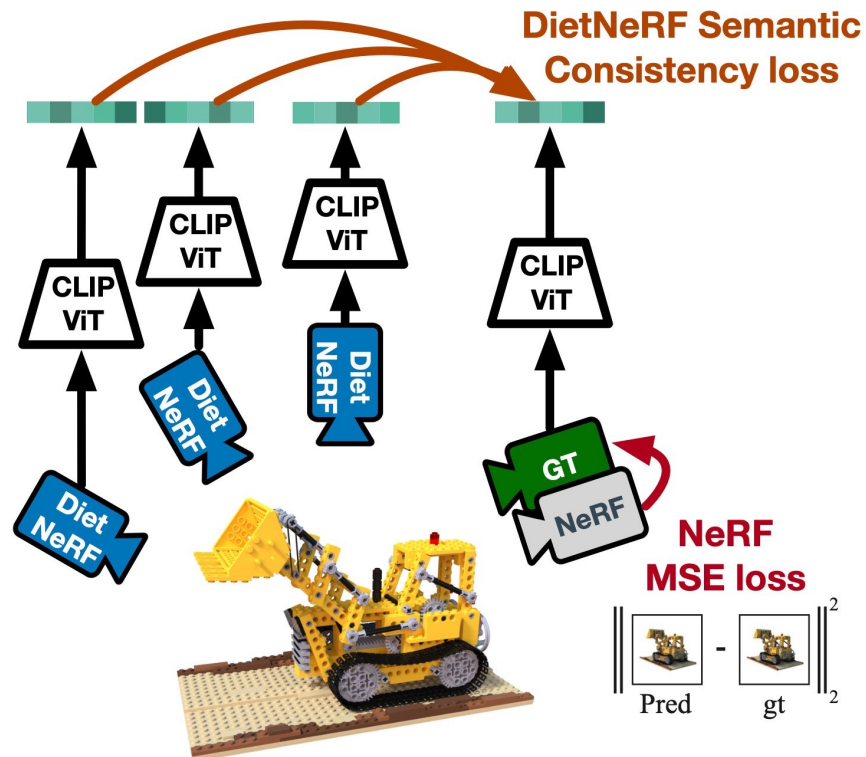
- CLIP is a **foundational model** trained on 400 million text-image pairs to connect **text** and **images**.
- CLIP **maps** texts and images to a latent **hypersphere**.
- **Image-text distance** can be measured as the distance over the hypersphere.



Few-Shot NeRF

DietNeRF

Use **priors** learned by a **text-image model**: CLIP

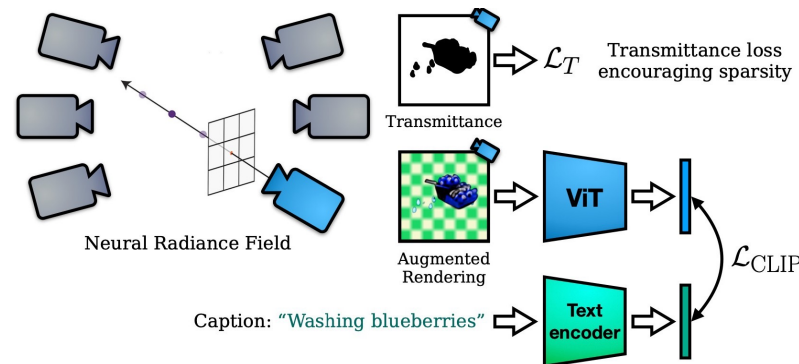


*“a bulldozer is a bulldozer
from any perspective”*

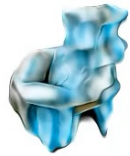
Extreme Case: Zero-Shot NeRF

- Let y be given **text prompt**.
- Perform gradient descent on the 3D representation θ using the following gradient:

$$\nabla_{\theta} \mathcal{L}_{\text{CLIP}}(\theta) = \frac{\partial}{\partial \theta} d(g(\theta; c_i) - y)$$



DreamFields



*The first 3D generative model
using 2D priors!*

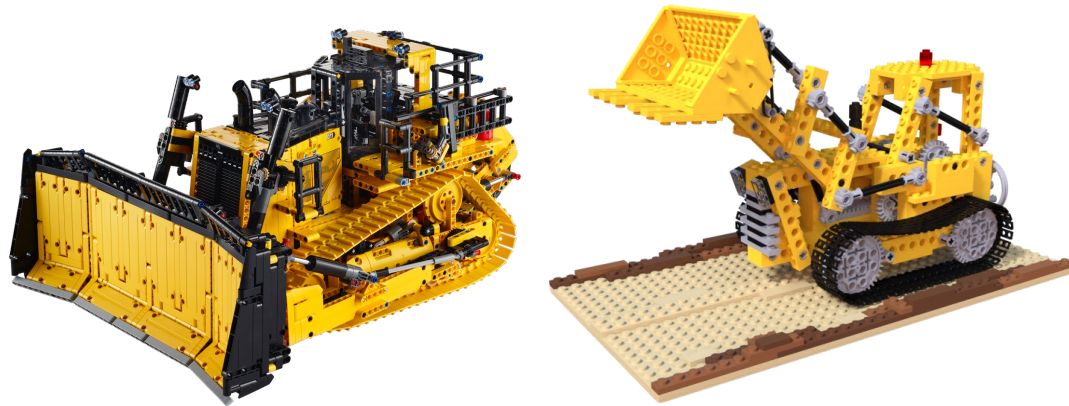
an archair in the shape of a _____.
an archair imitating a _.

a teapot in the shape of a _____.
a teapot imitating a _.

Can we use **pretrained image diffusion** models instead of CLIP?

Can we use **pretrained image diffusion** models instead of CLIP?

Q. 為什麼要用 diffusion ? 相比於 CLIP 有什麼優點 ?



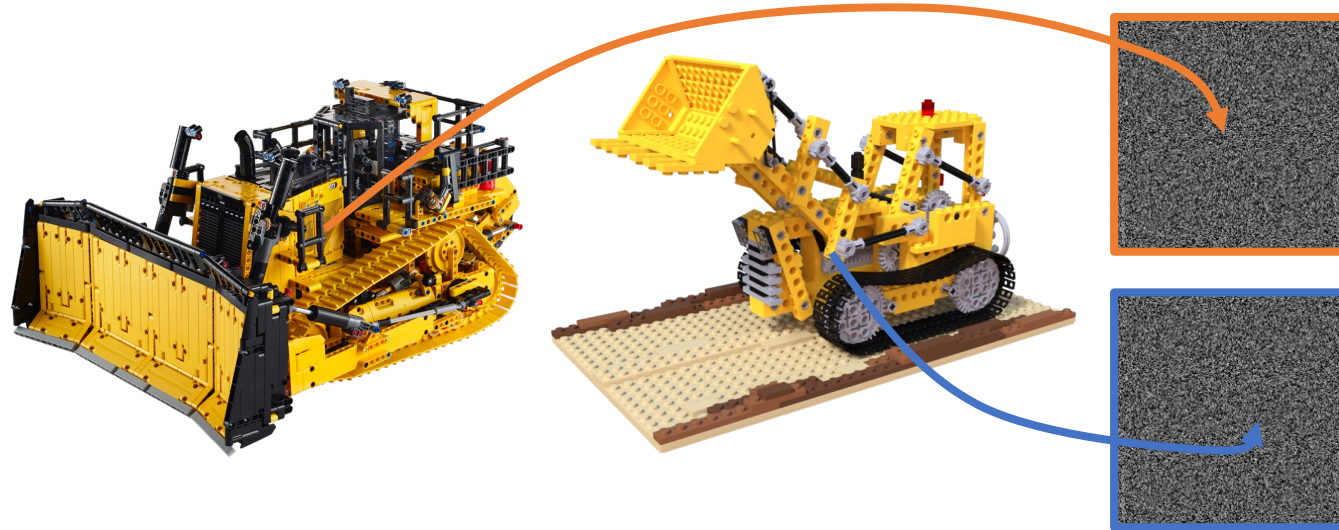
在 CLIP 的 semantic space
這兩張圖可能非常相近
因為他們都是樂高推土機

Can we use **pretrained image diffusion** models instead of CLIP?

Q. 為什麼要用 diffusion ? 相比於 CLIP 有什麼優點 ?

A.

CLIP 主要捕捉 high-level semantics , 缺乏幾何與細節的關注



在 diffusion 中
他們會對應到不同的 samples

Can we use **pretrained image diffusion** models instead of **CLIP**?

Q. 為什麼要用 diffusion ? 相比於 CLIP 有什麼優點 ?

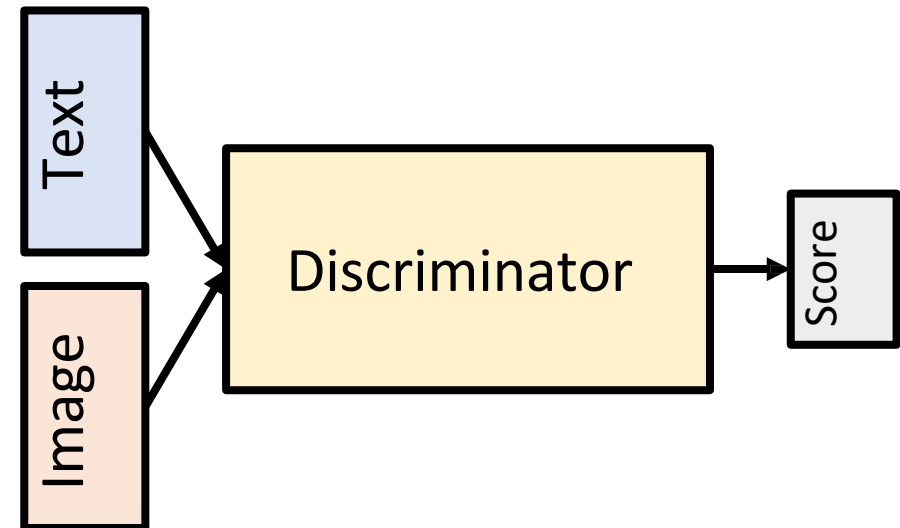
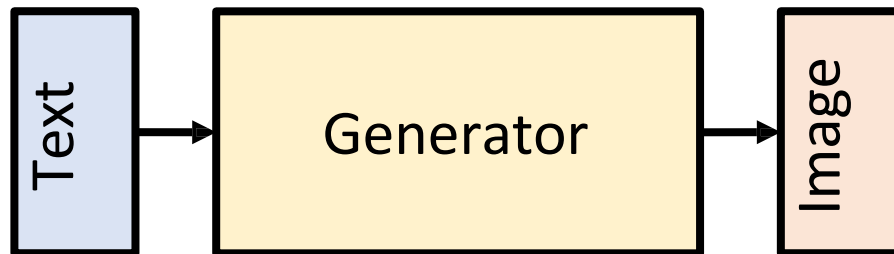
A.

CLIP 主要捕捉 high-level semantics , 缺乏幾何與細節的關注
Diffusion 能捕捉不同層次 , 從粗糙結構到精細紋理都有
且作為 generative model , diffusion 能產生更多樣的物體

Knowledge Distillation

Use a **generator** as a **discriminator**.

- **Generator**: Takes text as input and generates an image.
- **Discriminator**: Takes a pair of text and image as input and outputs the conformity.



Score Distillation Sampling (SDS)

Poole et al., DreamFusion: Text-to-3D using 2D Diffusion, ICLR 2023.

Score Distillation Sampling (SDS)

- How can we leverage a pretrained diffusion model to **measure the alignment** between the rendered image x_0 and the given **prompt y** ?
- See the loss function again:

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_{x_0 \sim q(x_0), t > 1, q(x_t | x_0)} [\|\hat{\epsilon}_\theta(x_t, y, t) - \epsilon_t\|^2] \\ &= \mathbb{E}_{x_0 \sim q(x_0), t > 1, q(x_t | x_0)} \left[\left\| \hat{\epsilon}_\theta \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t \right) - \epsilon_t \right\|^2 \right]\end{aligned}$$

Forward jump/前向跳躍

Score Distillation Sampling (SDS)

$$\mathcal{L} = \mathbb{E}_{x_0 \sim q(x_0), t > 1, q(x_t | x_0)} \left[\left\| \hat{\epsilon}_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t) - \epsilon_t \right\|^2 \right]$$

- If the training has converged, the loss for **real** data x_0 will be close to zero.
- Use the loss function as the measure of alignment.

3D Reconstruction via Optimization

For a random viewpoint c and a time step t , perform gradient descent on the 3D representation $\boldsymbol{\theta}$ using the following gradient:

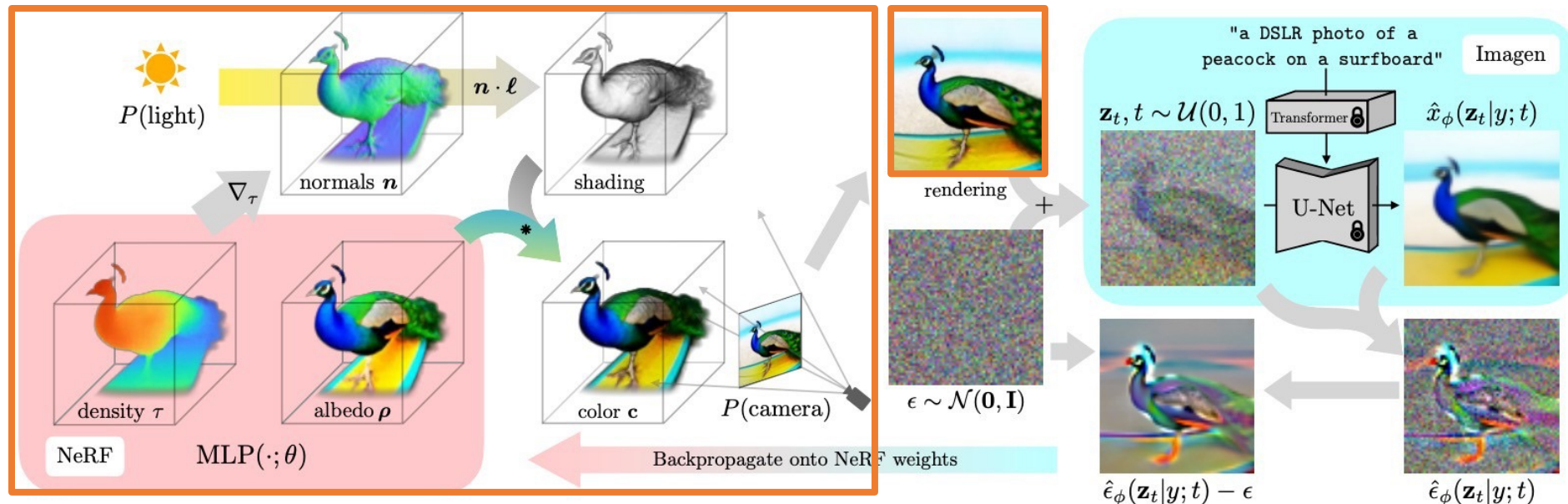
$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{DF}}(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \left\| \hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}} \left(\sqrt{\bar{\alpha}_t} g(\boldsymbol{\theta}; c) + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}_t, y, t \right) - \boldsymbol{\varepsilon}_t \right\|^2$$

where $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a random sample from a standard normal distribution.

Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta}(\underbrace{\sqrt{\bar{\alpha}_t} g(\theta; c)}_{x_0} + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t) - \epsilon_t \right\|^2$$

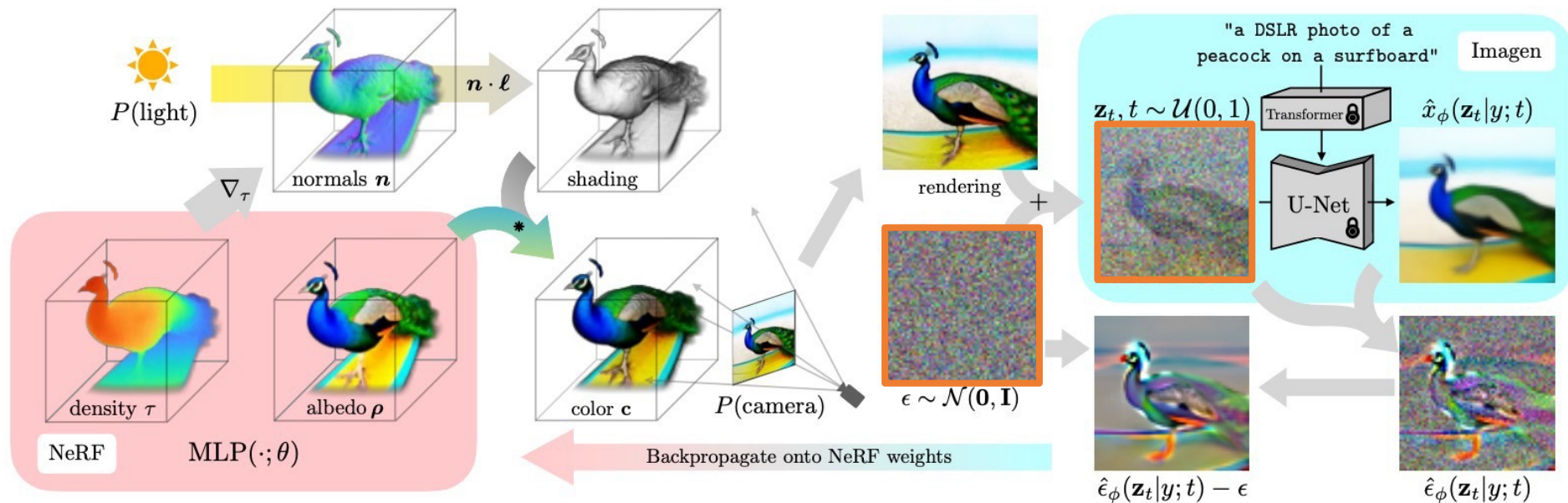
1. **Render** the 3D representation θ into a specific view c .



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\underbrace{\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t}_{x_t}, y, t \right) - \epsilon_t \right\|^2$$

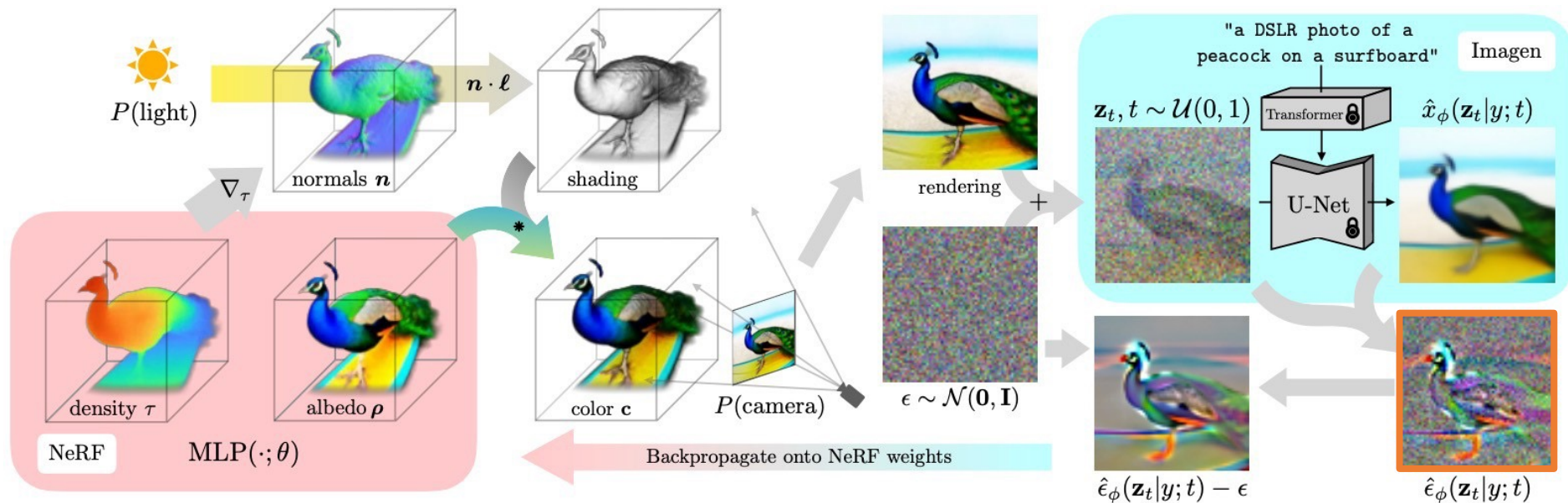
2. Add noise to the rendered image $g(\theta; c)$.



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t \right) - \epsilon_t \right\|^2$$

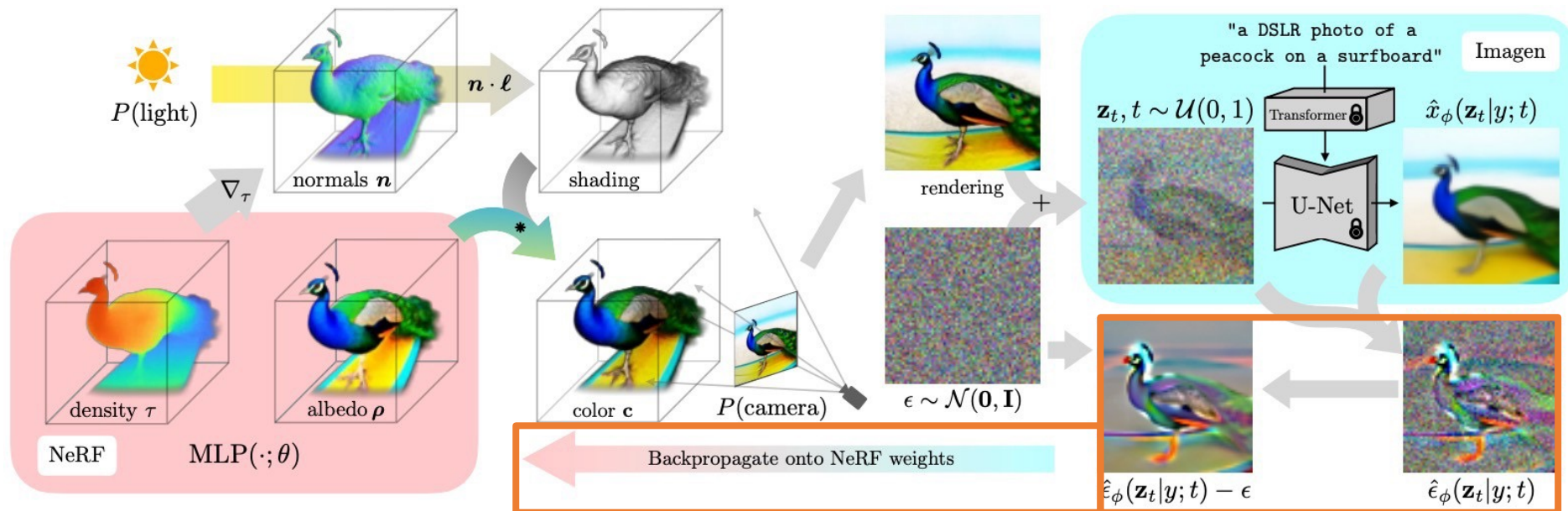
3. Predict the noise using the noise predictor.



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t \right) - \epsilon_t \right\|^2$$

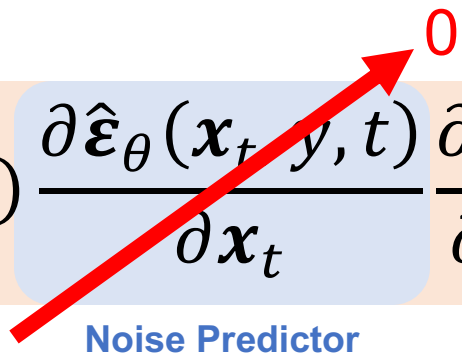
4. Backpropagate onto θ while minimizing the ϵ_t difference.



Score Distillation Sampling (SDS)

$$\text{Let } \mathbf{x}_t = \sqrt{\bar{\alpha}_t} g(\boldsymbol{\theta}; c) + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}_t$$

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{DF}}(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \|\hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t) - \boldsymbol{\varepsilon}_t\|^2$$


$$= 2(\hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t) - \boldsymbol{\varepsilon}_t) \underbrace{\frac{\partial \hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t)}{\partial \mathbf{x}_t}}_{\text{Noise Predictor Jacobian}} \frac{\partial \mathbf{x}_t}{\partial \boldsymbol{\theta}} \quad \text{Chain Rule}$$

原本沒有忽略 noise predictor Jacobian 計算的叫做 denoising score matching (**DSM**) loss

Score Distillation Sampling (SDS)

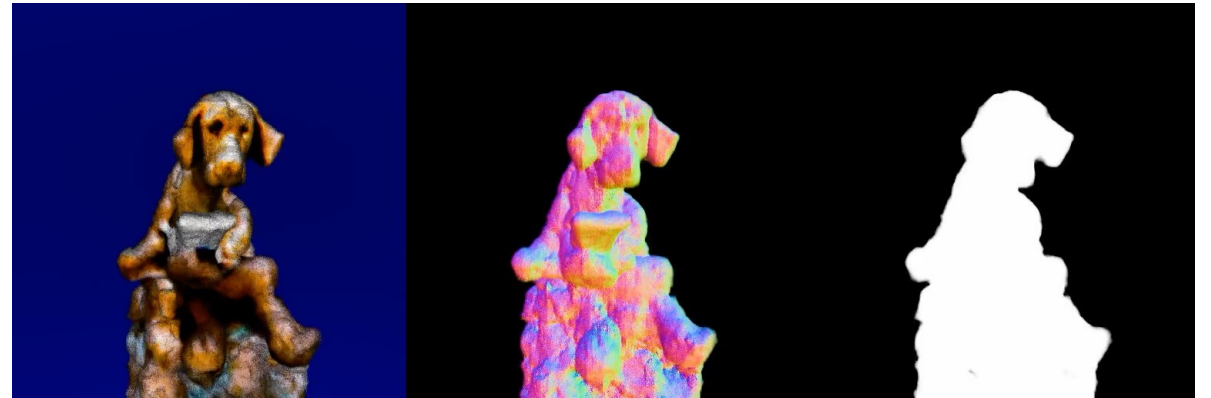
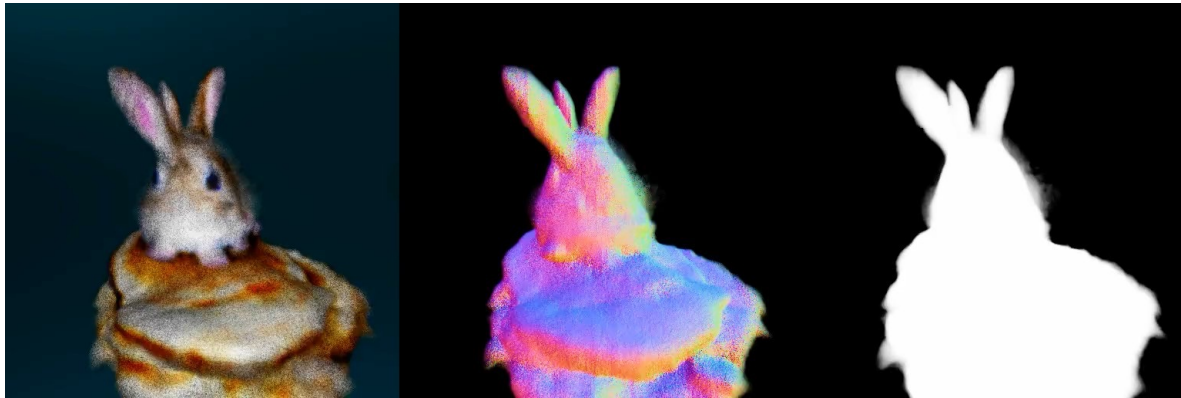
- Drop the noise predictor Jacobian to save computation time and memory.
- The final SDS gradient is thus defined as follows:

$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\theta) = (\hat{\epsilon}_{\theta}(\mathbf{x}_t, y, t) - \epsilon_t) \frac{\partial \mathbf{x}_t}{\partial \theta}$$

Score Distillation Sampling (SDS)

Effectiveness of dropping the noise predictor Jacobian.

w/o Jacobian



Score Distillation Sampling (SDS)

Effectiveness of dropping the noise predictor Jacobian.

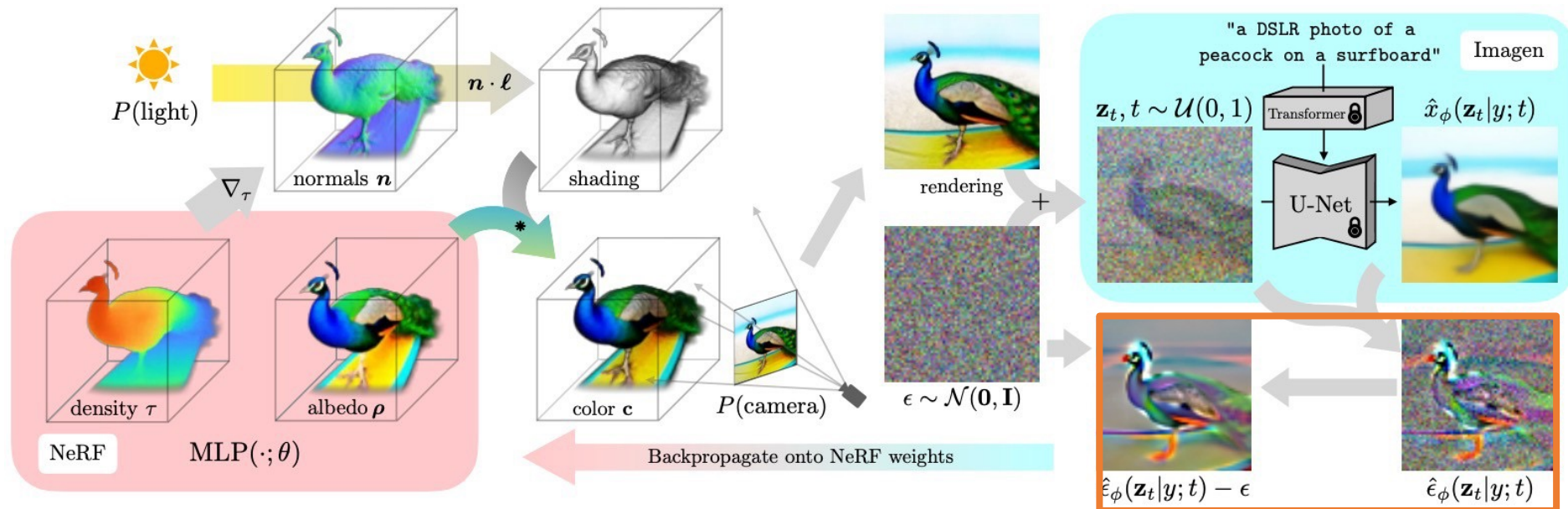
w/ Jacobian



*It takes a similar amount of computation time
but requires twice the VRAM memory.*

Why is it Score Distillation Sampling (SDS)?

We **distill** the knowledge learned by a pretrained diffusion model in the form of a **score** (noise ϵ_t).



Why SDS Instead of Reverse Diffusion?

- This is a scenario where the data (in this case, images) are **parameterized differently from how they were represented** during the training of the diffusion model.
- Hence, instead of using the reverse process, we **sample** the outputs by **iteratively applying gradient descent** from random samples.

DreamFusion Results



"frog wearing a sweater"



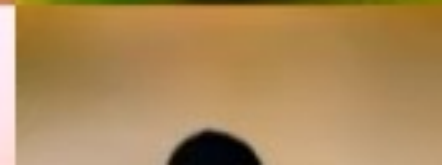
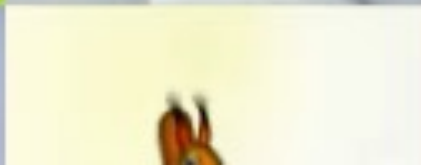
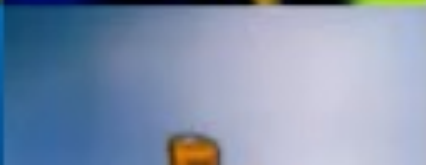
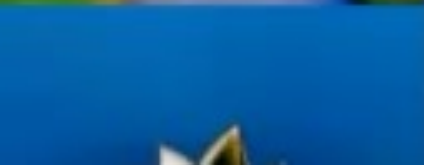
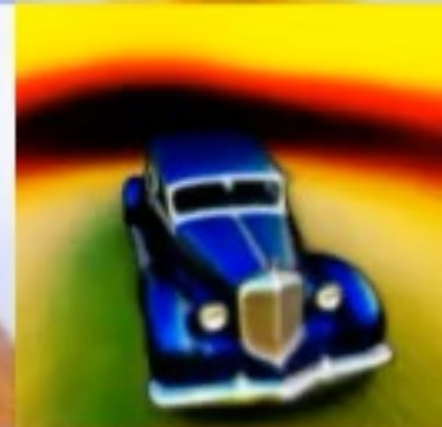
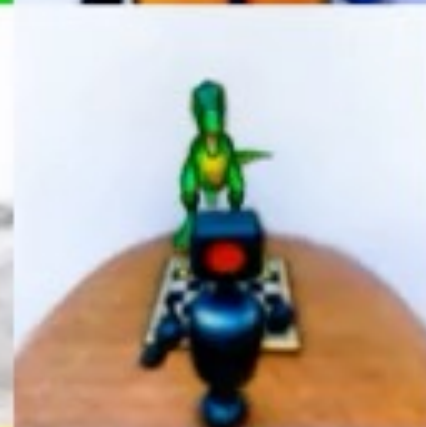
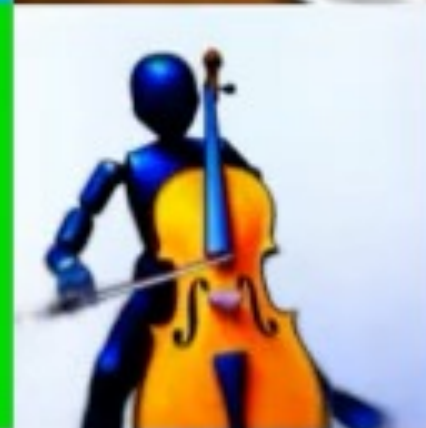
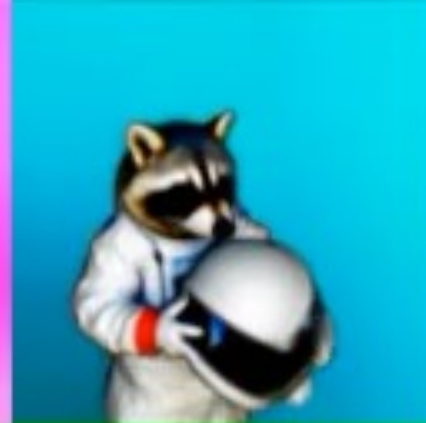
"eggshell broken in two
with an adorable chick
standing next to it"



"ghost eating a hamburger"



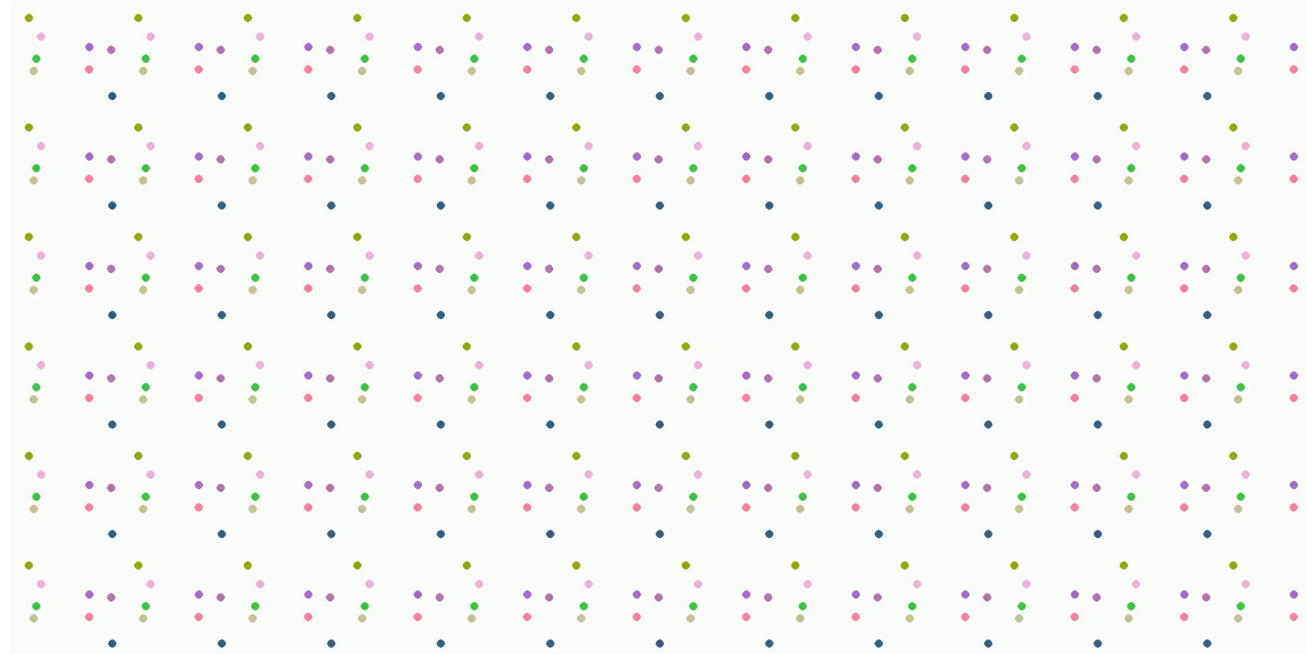
"a pig wearing a backpack"



SDS Applications

SDS can be applied to generate various types of visual content that can be rendered into images.

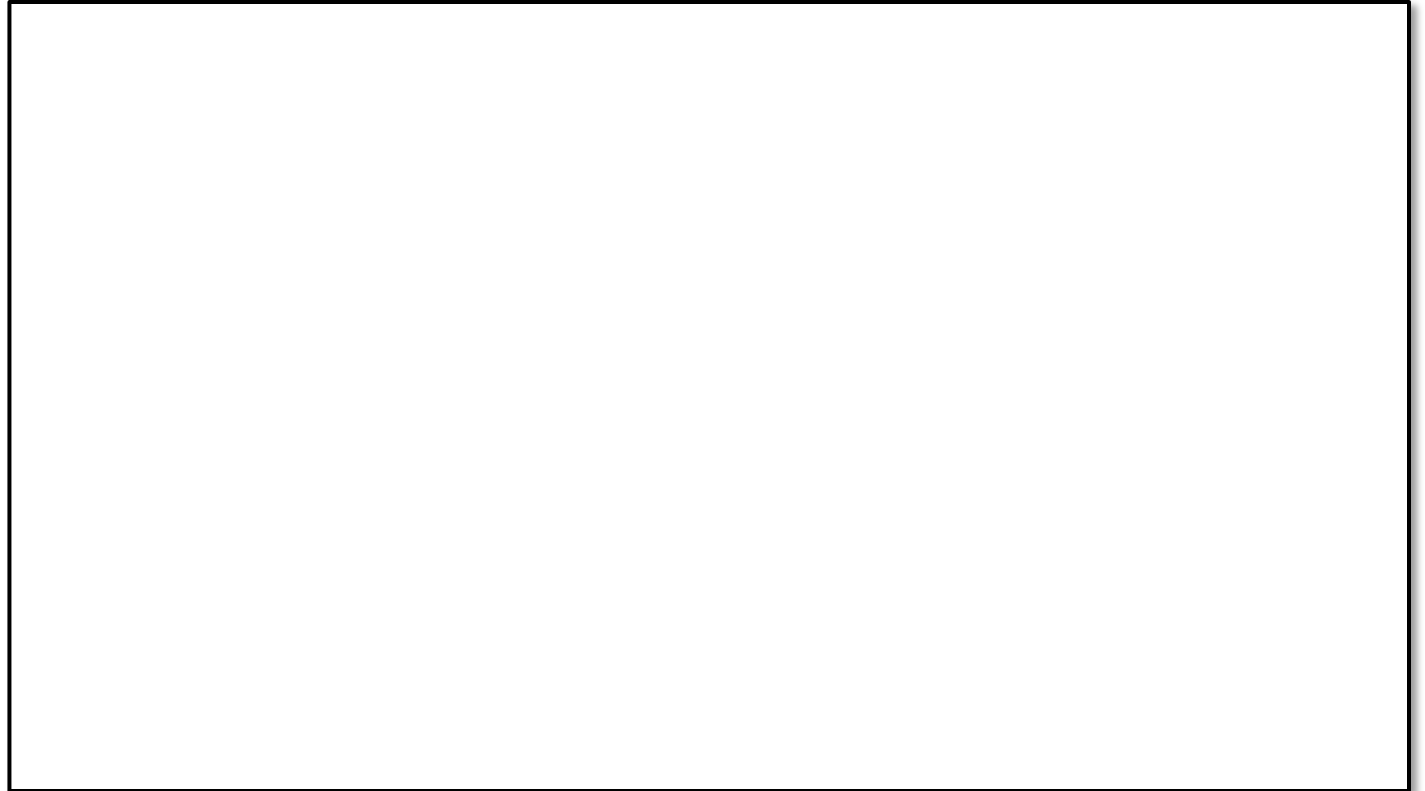
E.g., **vector images**.



SDS Applications

SDS can be applied to generate various types of visual content that can be rendered into images.

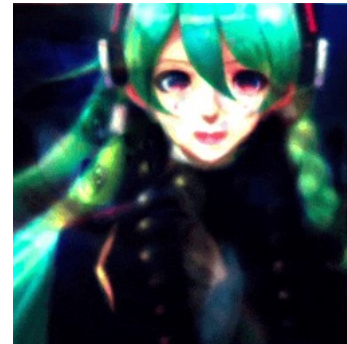
E.g., **word arts**.



SDS Applications

SDS can be applied to generate various types of visual content that can be rendered into images.

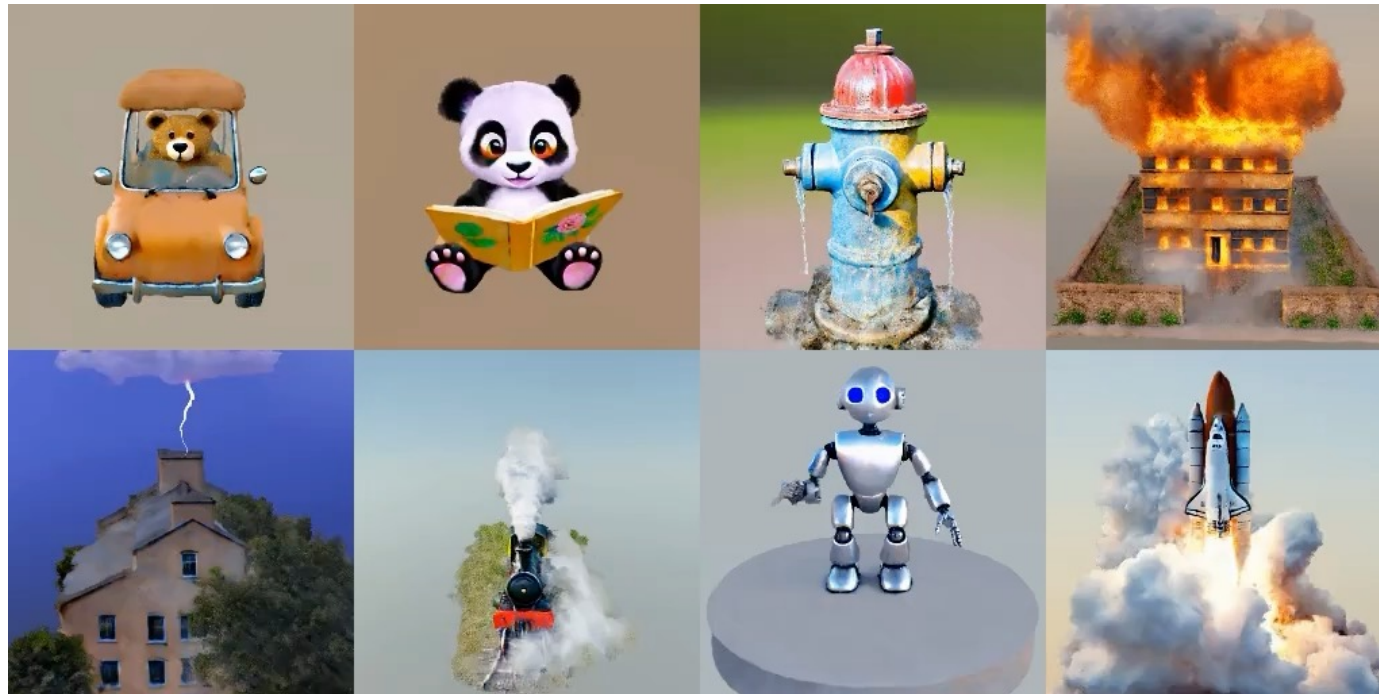
E.g., **illusions**.



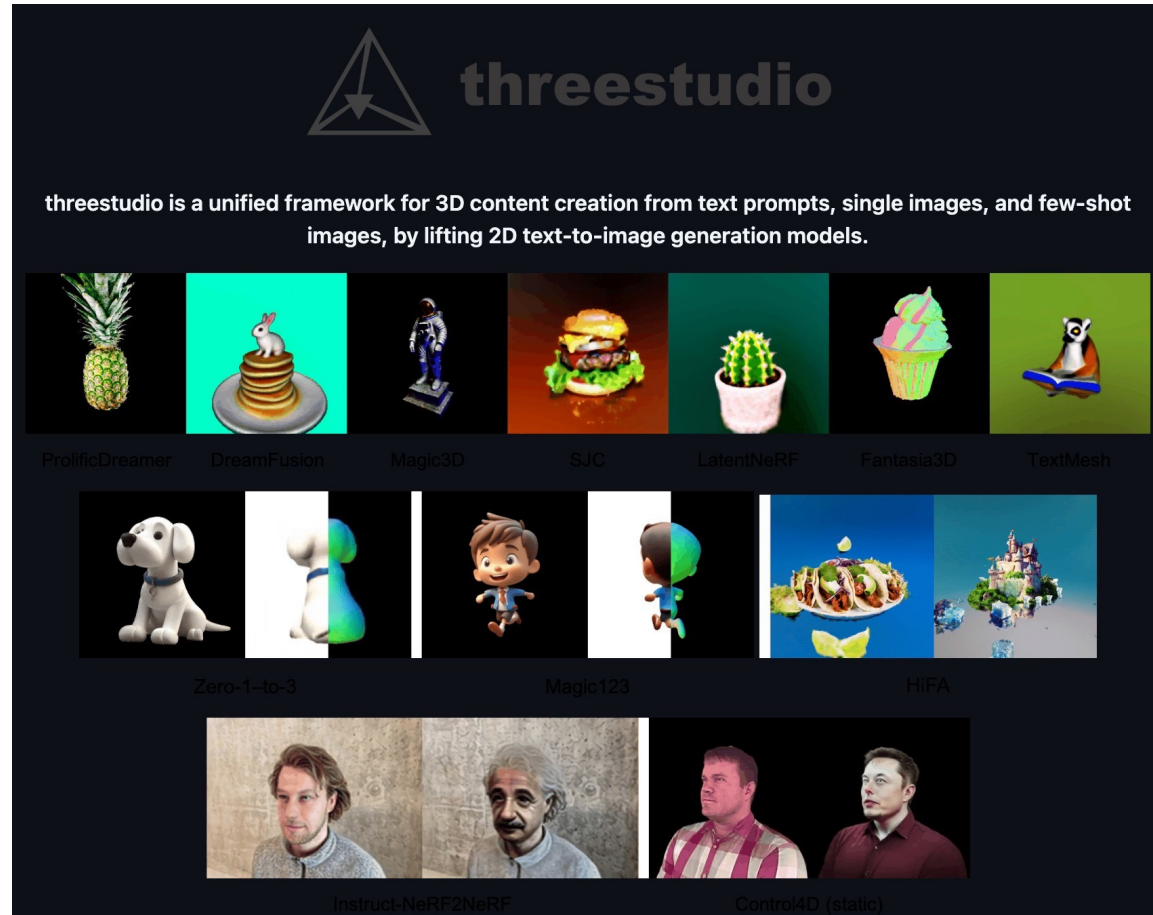
SDS Applications

SDS can be applied to generate various types of visual content that can be rendered into images.

E.g., 4D (3D + animation)



Stable-DreamFusion



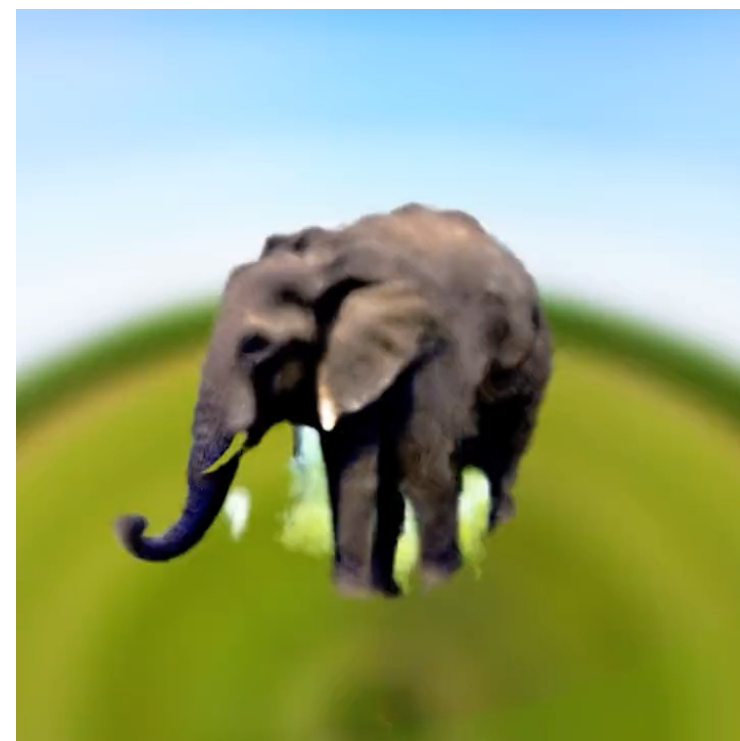
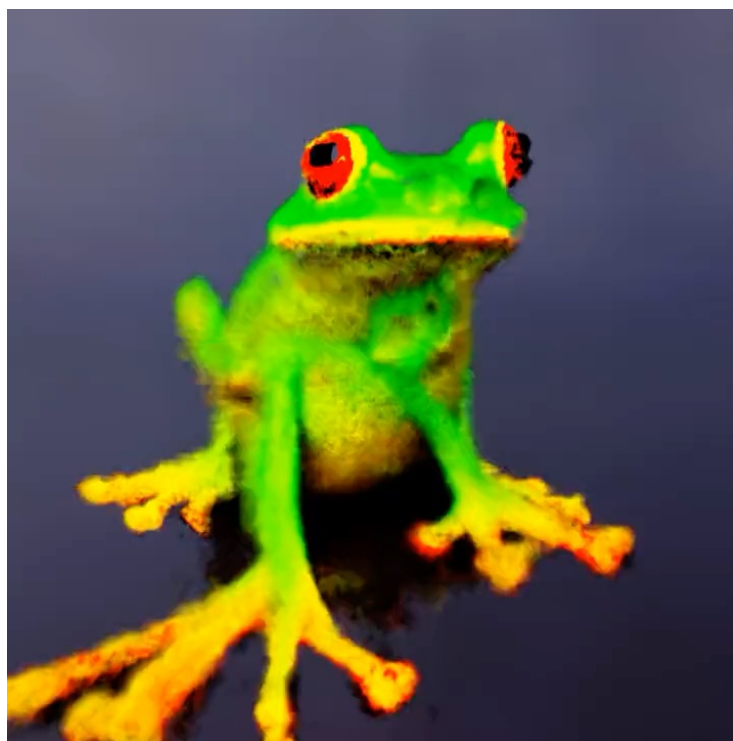
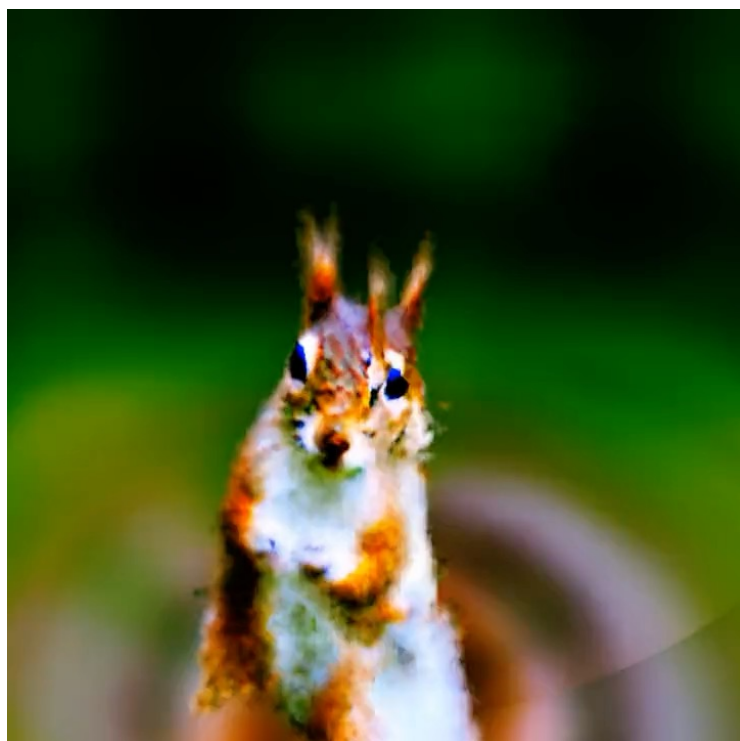
Limitation of 3D Generation from 2D Priors

Janus Problem

→ 雅努斯
羅馬神話中的兩面之神



不是他

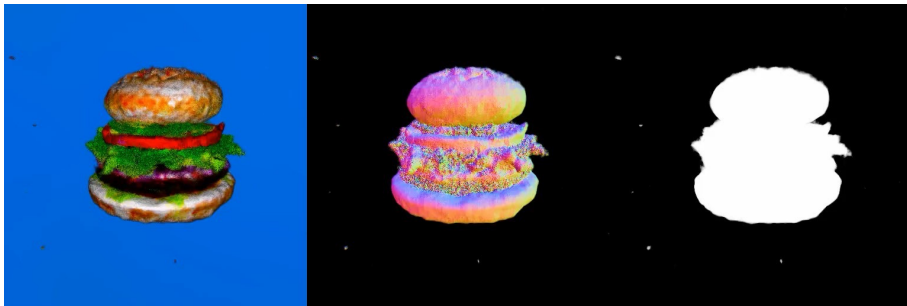


Limitation of SDS

- It does not converge well without a high CFG weight.
- With a high CFG weight, it loses diversity.

“a delicious hamburger”

$w = 400$



$w = 7.5$



NeRF Initial.



Seed=0

Seed=1

Seed=2

Seed=3

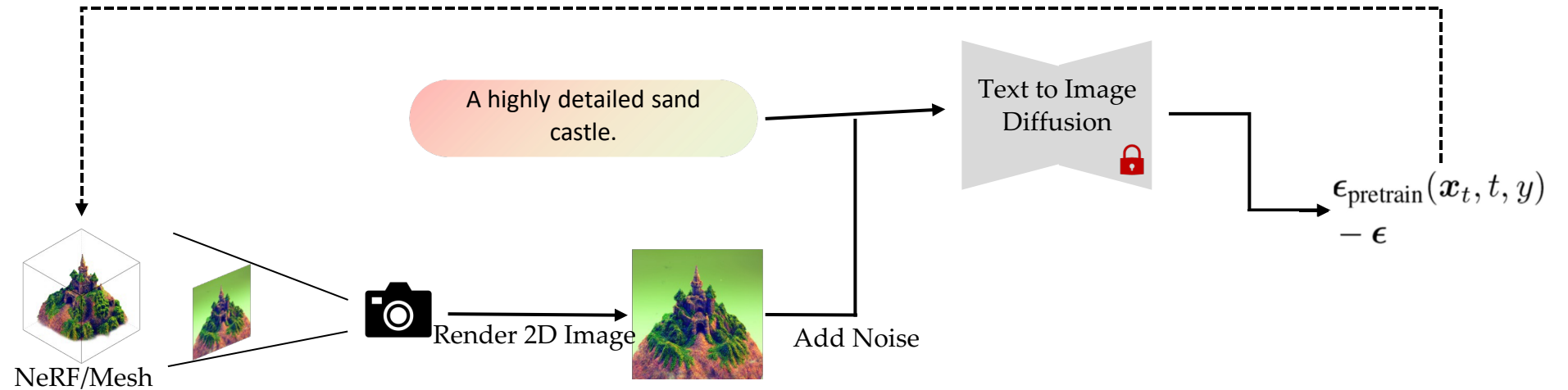
“gingerbread man”

Improving SDS

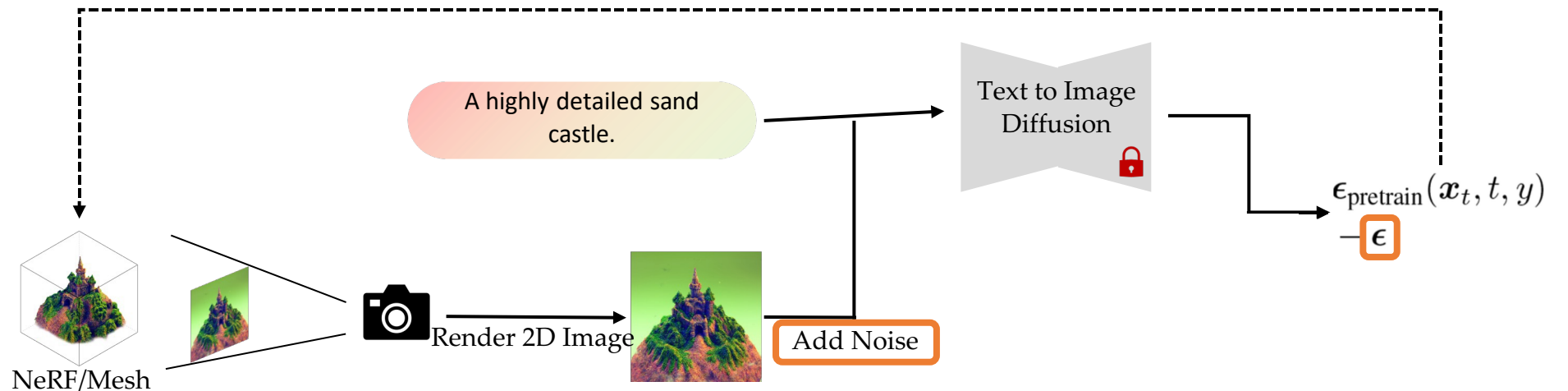
$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\theta) = (\hat{\epsilon}_{\theta}(\mathbf{x}_t, y, t) - \epsilon_t) \frac{\partial \mathbf{x}_t}{\partial \theta}$$

- ϵ_t is a random sample from a standard normal distribution.
- What can we use instead of the random sample?

Score Distillation Sampling (SDS)



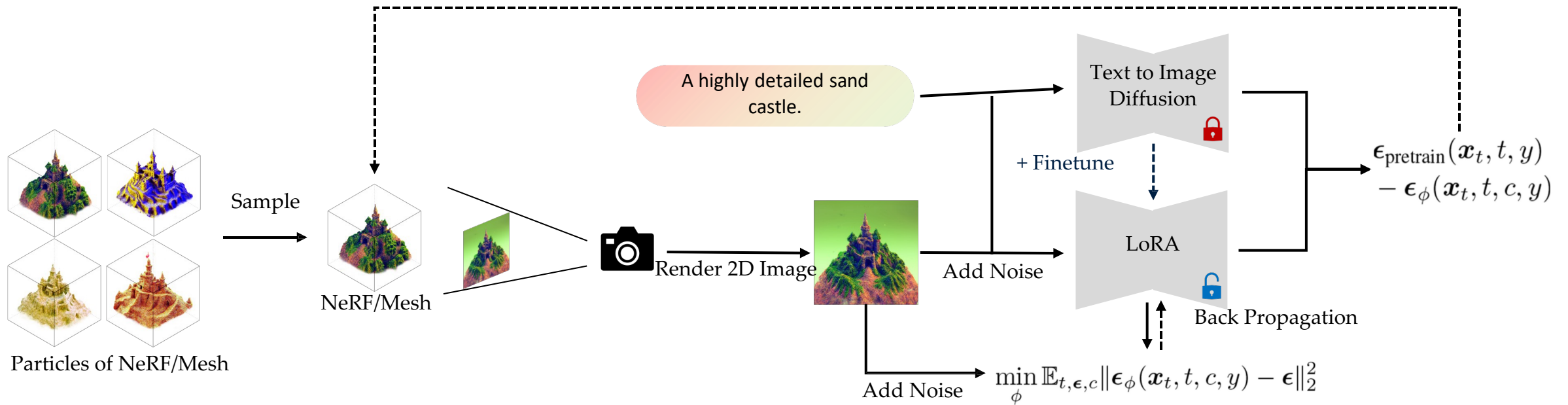
Score Distillation Sampling (SDS)



這隨便加的 random noise 真的是最能代表當前 3D 場景分布的 noise 嗎？

Variational Score Distillation (VSD)

Finetune the score prediction network with **LoRA** while reconstructing **multiple** 3D representations with the same text prompt.



Variational Score Distillation (VSD)

$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\theta) = (\hat{\epsilon}_{\theta}(\mathbf{x}_t, y, t) - \epsilon_t) \frac{\partial \mathbf{x}_t}{\partial \theta}$$

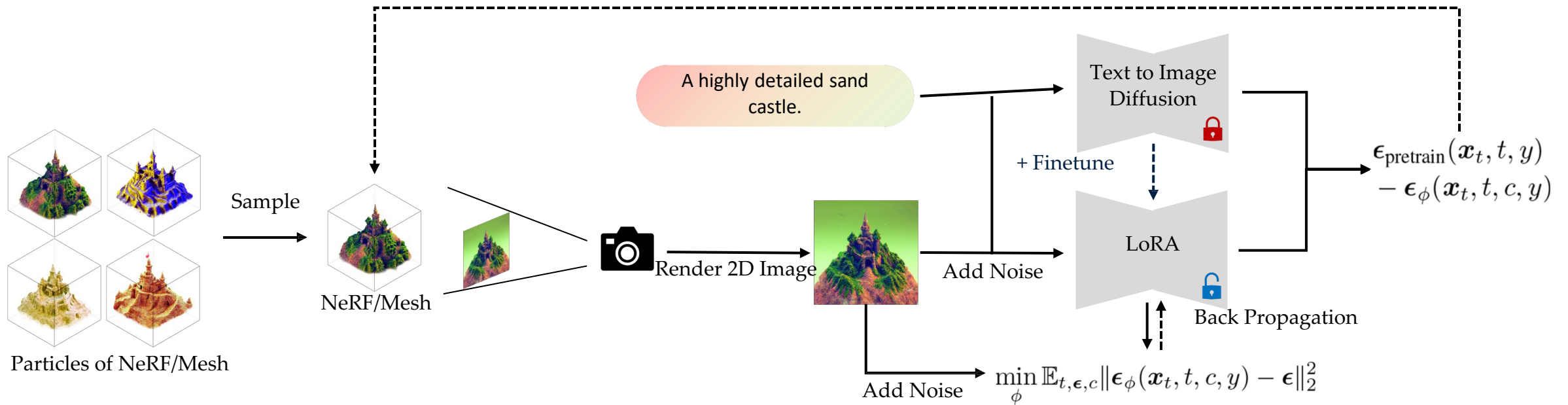


$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\theta) = (\hat{\epsilon}_{\theta}(\mathbf{x}_t, y, t) - \hat{\epsilon}_{\text{LoRA}}(\mathbf{x}_t, y, t)) \frac{\partial \mathbf{x}_t}{\partial \theta}$$

Update the LoRA parameters and 3D representations alternatively.

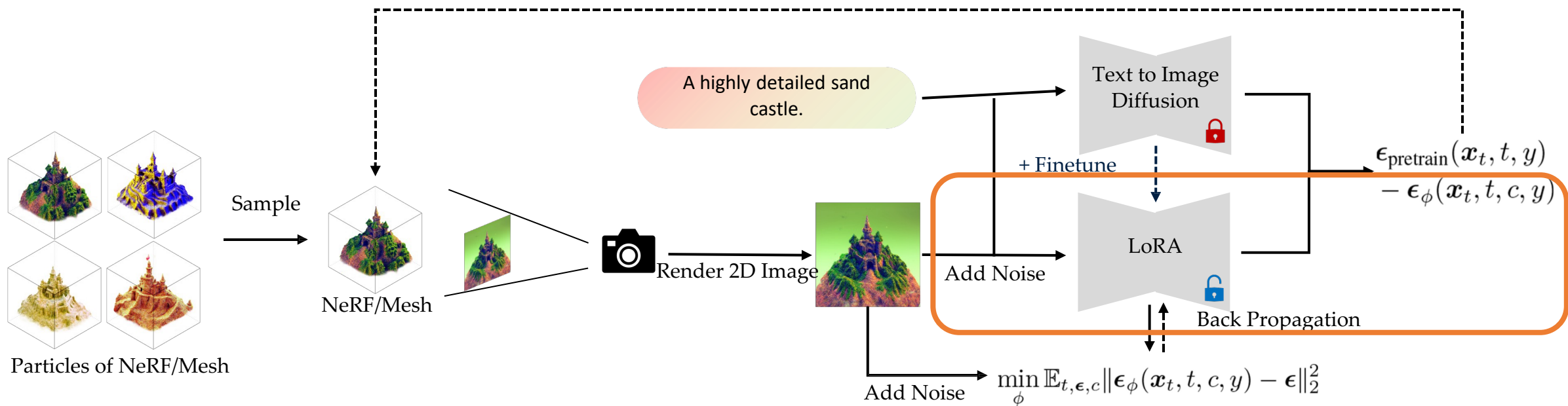
Variational Score Distillation (VSD)

Finetune the score prediction network with **LoRA** while reconstructing **multiple** 3D representations with the same text prompt.



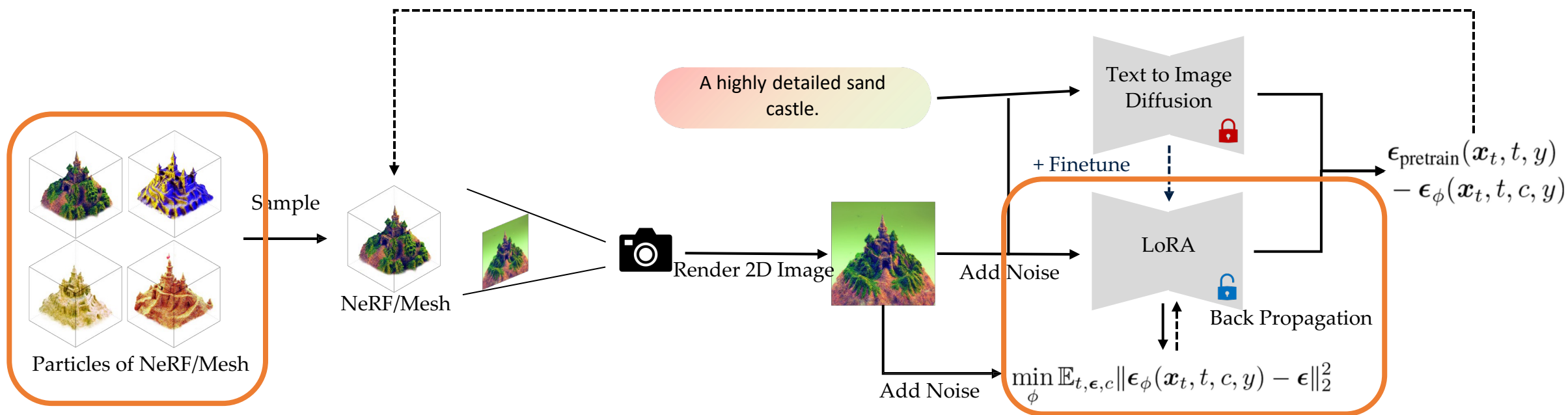
Variational Score Distillation (VSD)

Noise 改成是用 LoRA 學出來的



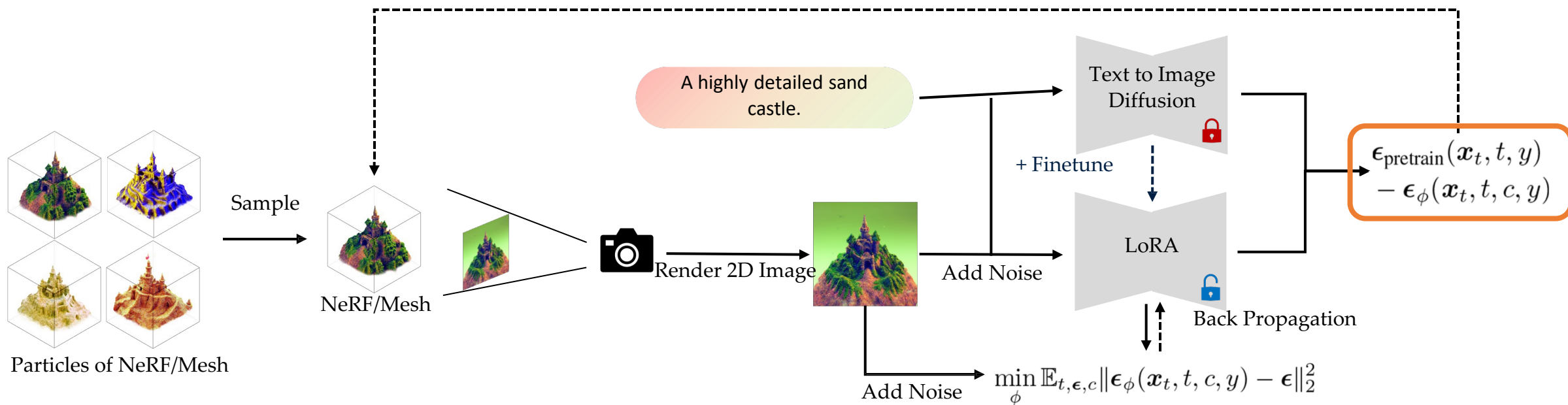
Variational Score Distillation (VSD)

透過多個 NeRF 的 samples 來讓 LoRA 學到當前場景的 distribution



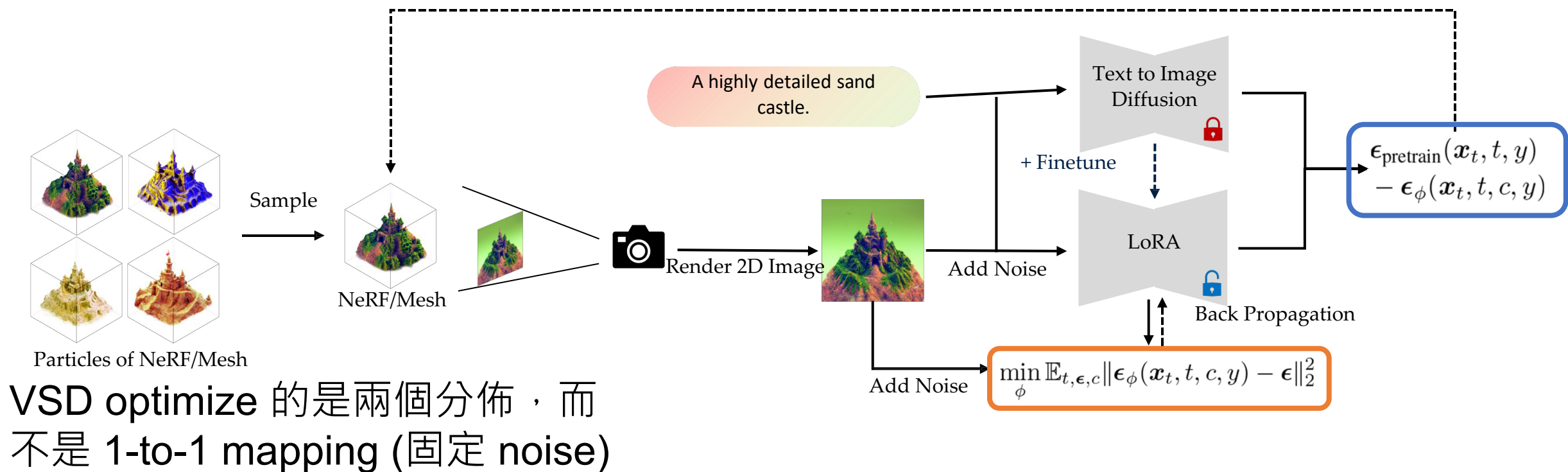
Variational Score Distillation (VSD)

更新 NeRF 的部分相同，不過 noise target 變成 LoRA-finetuned diffusion 的 noise prediction



Variational Score Distillation (VSD)

Iteratively optimize **LoRA** & **NeRF**



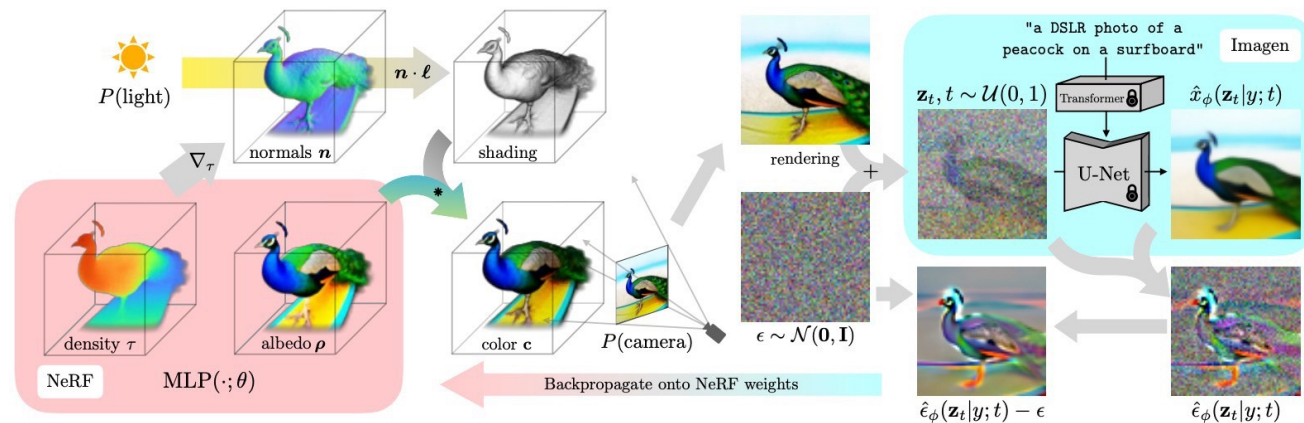
References

- [ICLR'23] SDS (Score Distillation Sampling)
- [ICCV'23] DDS (Delta Denoising Score)
- [NeurIPS'23] VSD (Variational Score Distillation)
- [ECCV'24] ADD (Adversarial Diffusion Distillation)
- [CVPR'24] DMD (Distribution Matching Distillation)
- [SIGGRAPH Asia'24] LADD (Latent Adversarial Diffusion Distillation)
- [NeurIPS'24] DMD2 (Improved Distribution Matching Distillation)
- [NeurIPS'24] SDI (Score Distillation via Reparametrized DDIM)

3D Generation via SDS

- Let y be given **text prompt**.
- Perform gradient descent on the 3D representation θ using the following gradient:

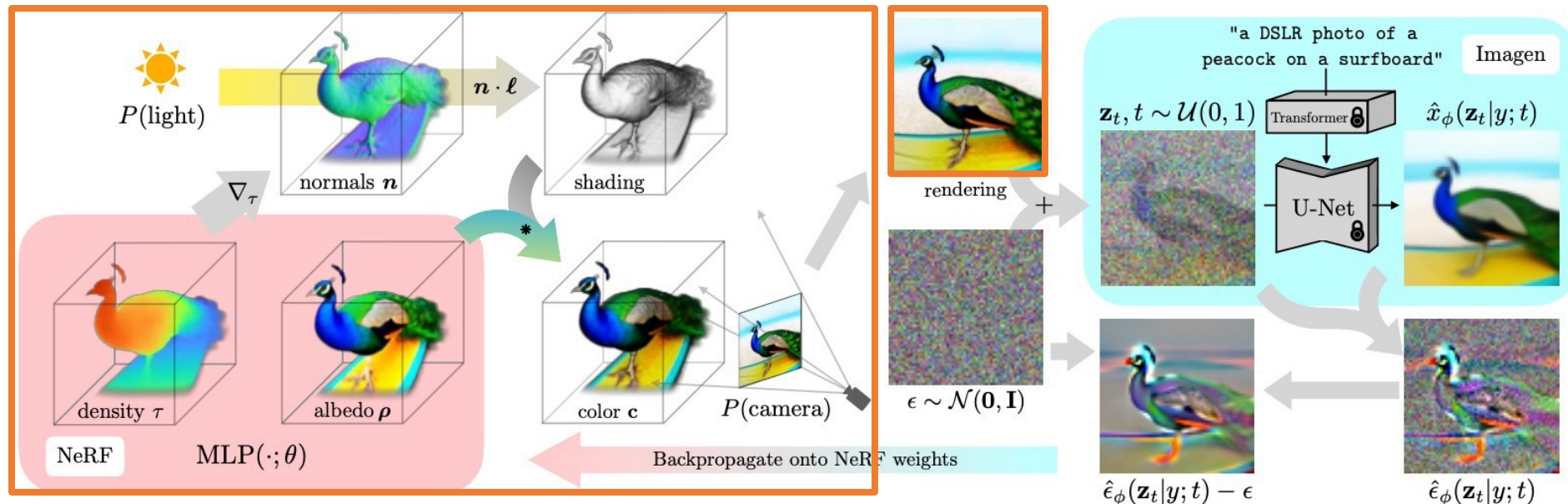
$$\nabla_{\theta} \mathcal{L}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t \right) - \epsilon_t \right\|^2$$



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta}(\underbrace{\sqrt{\bar{\alpha}_t} g(\theta; c)}_{x_0} + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t) - \epsilon_t \right\|^2$$

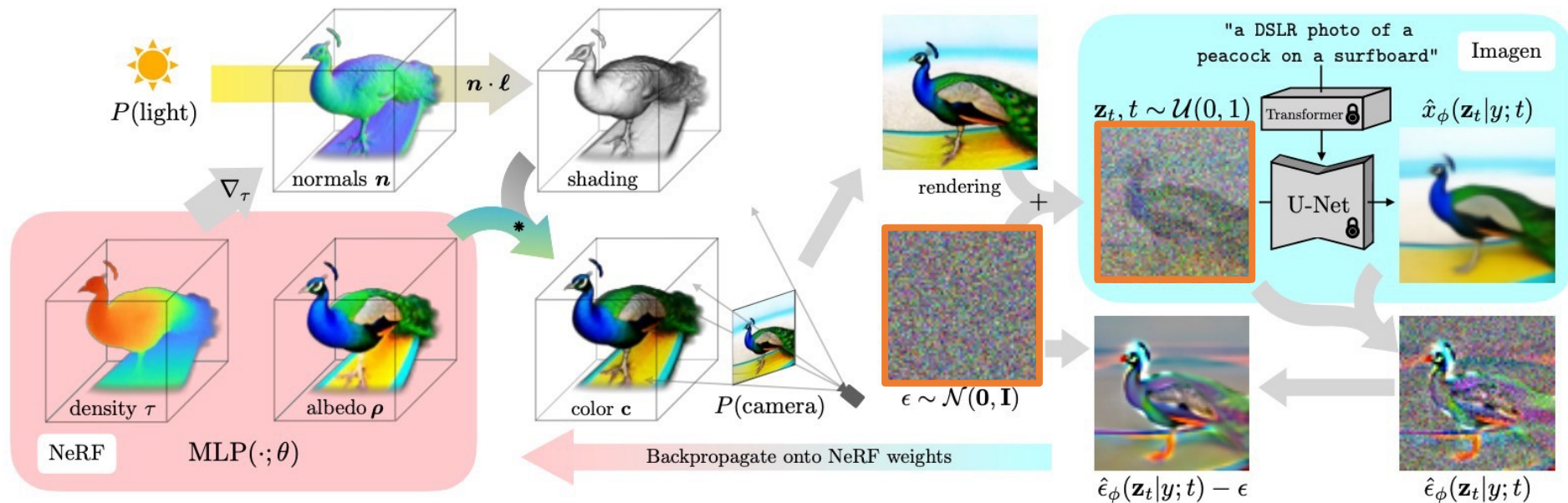
1. **Render** the 3D representation θ into a specific view c .



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\underbrace{\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t}_{x_t}, y, t \right) - \epsilon_t \right\|^2$$

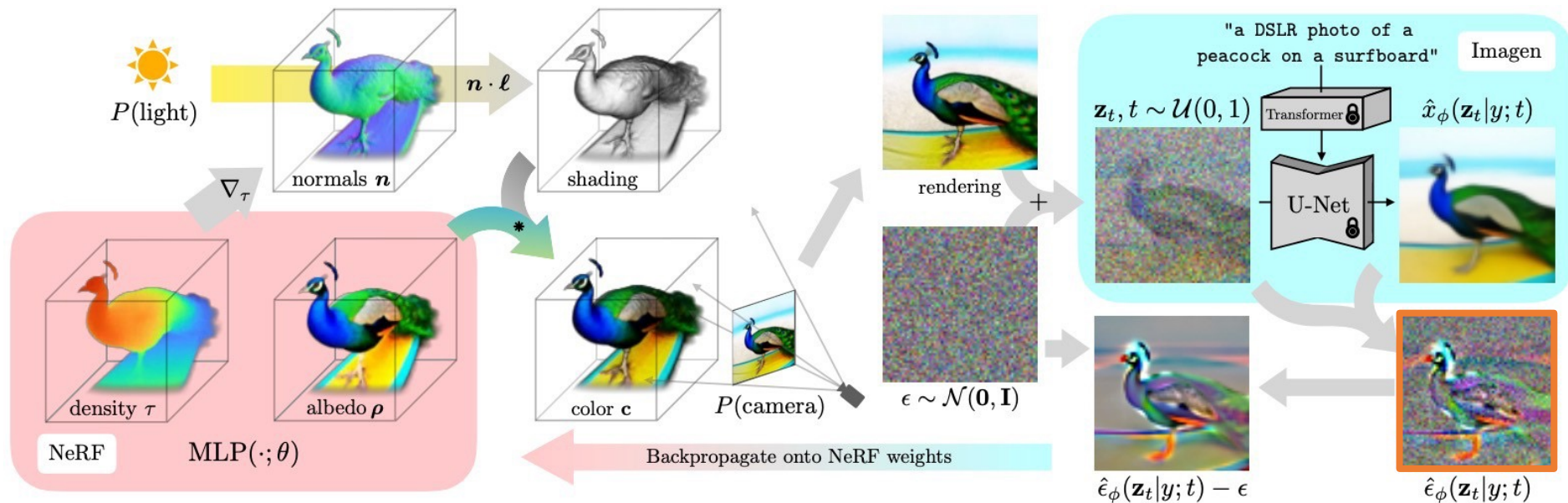
2. Add noise to the rendered image $g(\theta; c)$.



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t \right) - \epsilon_t \right\|^2$$

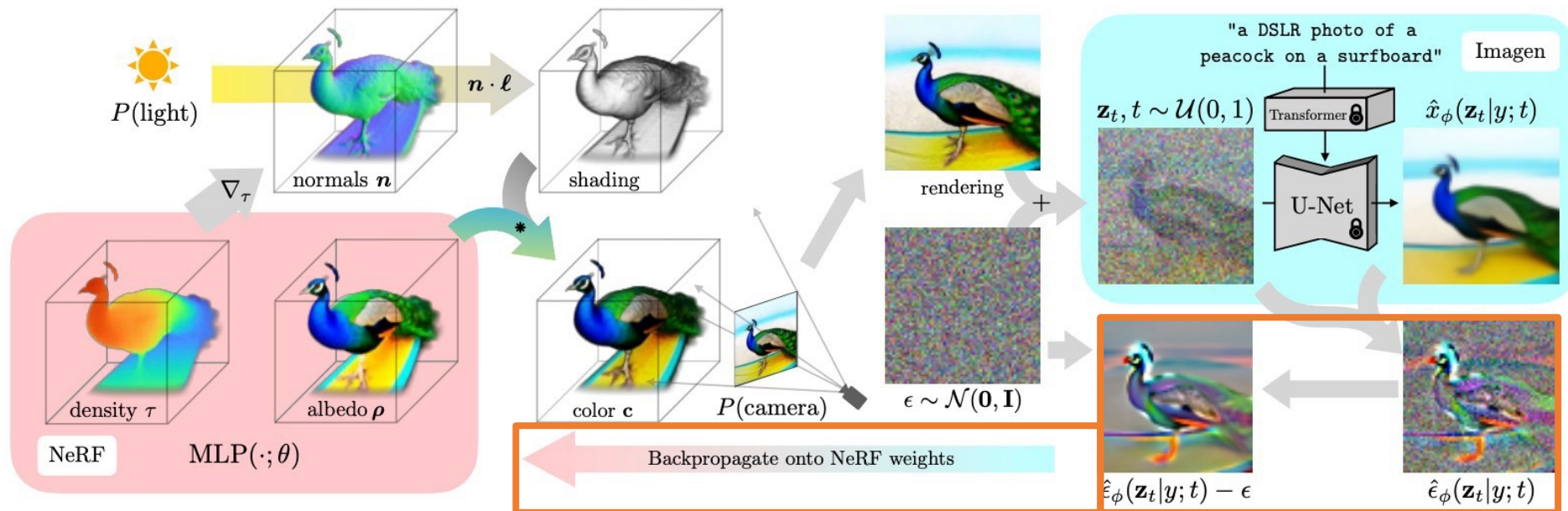
3. Predict the noise using the noise predictor.



Score Distillation Sampling (SDS)

$$\nabla_{\theta} \mathcal{L}_{\text{DF}}(\theta) = \frac{\partial}{\partial \theta} \left\| \hat{\epsilon}_{\theta} \left(\sqrt{\bar{\alpha}_t} g(\theta; c) + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, y, t \right) - \epsilon_t \right\|^2$$

4. Backpropagate onto θ while minimizing the ϵ_t difference.



DreamFusion Results



"frog wearing a sweater"



"eggshell broken in two
with an adorable chick
standing next to it"



"ghost eating a hamburger"



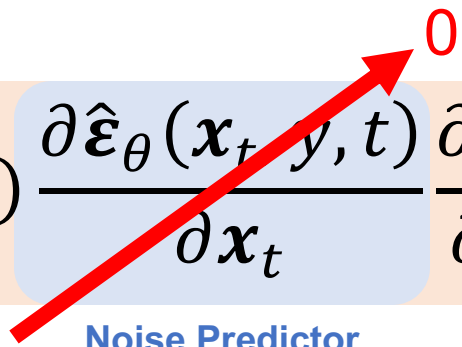
"a pig wearing a backpack"

Score Distillation Sampling (SDS)

$$\text{Let } \mathbf{x}_t = \sqrt{\bar{\alpha}_t} g(\boldsymbol{\theta}; c) + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}_t$$

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{DF}}(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \|\hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t) - \boldsymbol{\varepsilon}_t\|^2$$

$$= 2(\hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t) - \boldsymbol{\varepsilon}_t) \frac{\partial \hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t)}{\partial \mathbf{x}_t} \frac{\partial \mathbf{x}_t}{\partial \boldsymbol{\theta}} \quad \text{Chain Rule}$$

A red arrow points from the text "Noise Predictor Jacobian" to the fraction $\frac{\partial \hat{\boldsymbol{\varepsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, y, t)}{\partial \mathbf{x}_t}$ in the equation above.

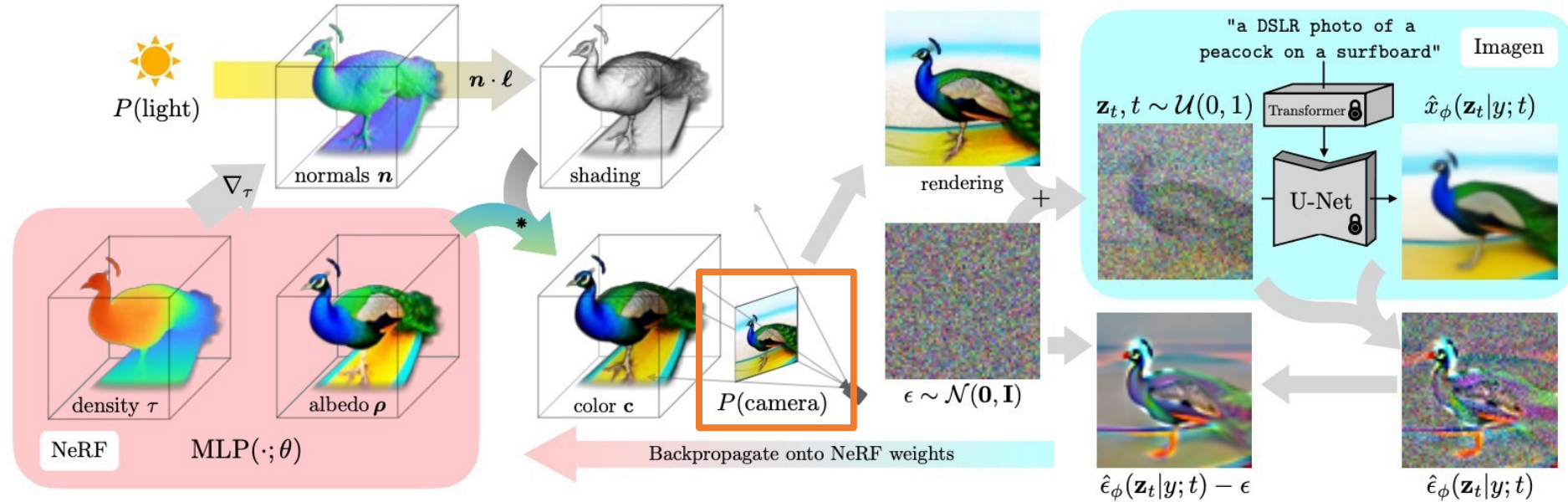
Noise Predictor
Jacobian

Score Distillation Sampling

$$\mathcal{L}_{\text{SDS}}(\mathbf{x}_0) = \mathbb{E}_{t, \epsilon} \left[\|\epsilon_{\phi}(\mathbf{x}_t, y, t) - \epsilon\|_2^2 \right]$$

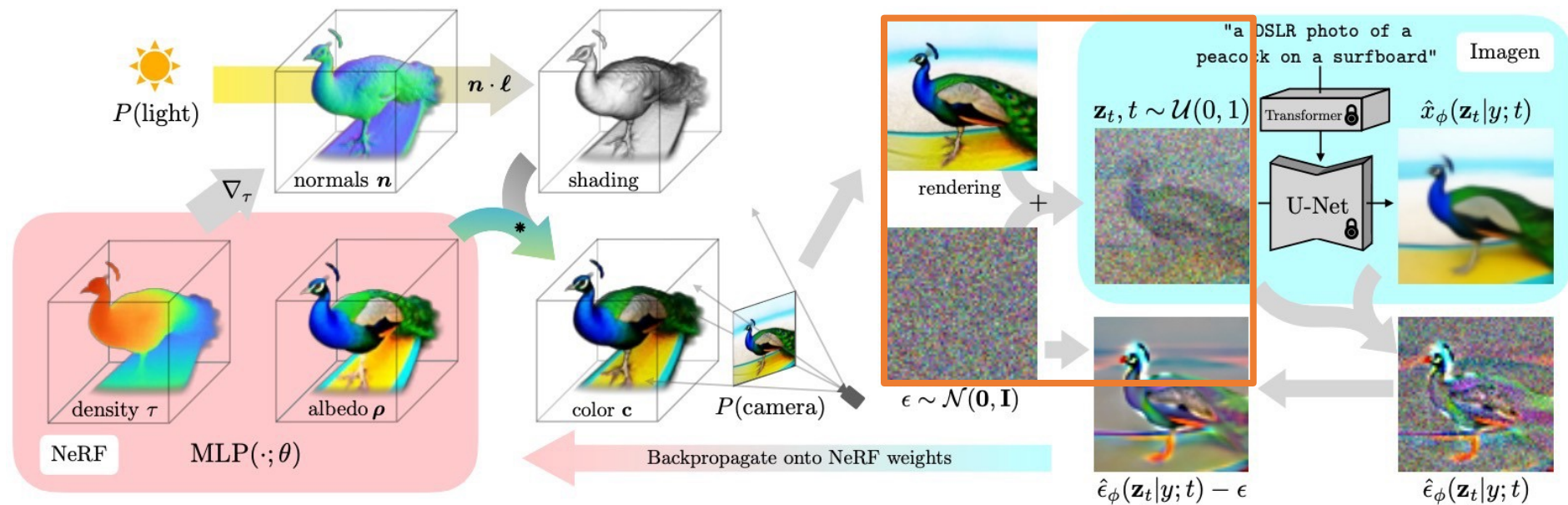
- $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$: Noisy image.
- $\mathbf{x}_0$: Input image (NOT parameterized as pixel colors).
- y : Input text prompt.
- $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$: Random noise added to the input image.

Score Distillation Sampling (SDS) 的直觀解釋



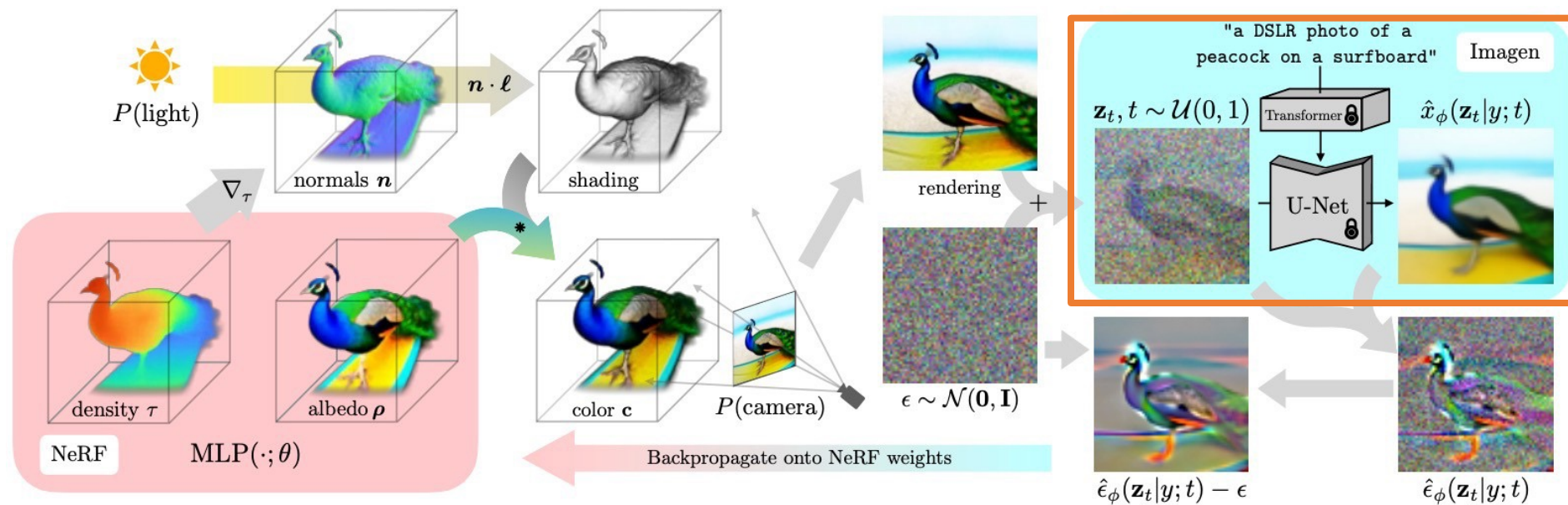
這邊先有一張爛爛的影像
看起來可能完全不像真實影像

Score Distillation Sampling (SDS) 的直觀解釋



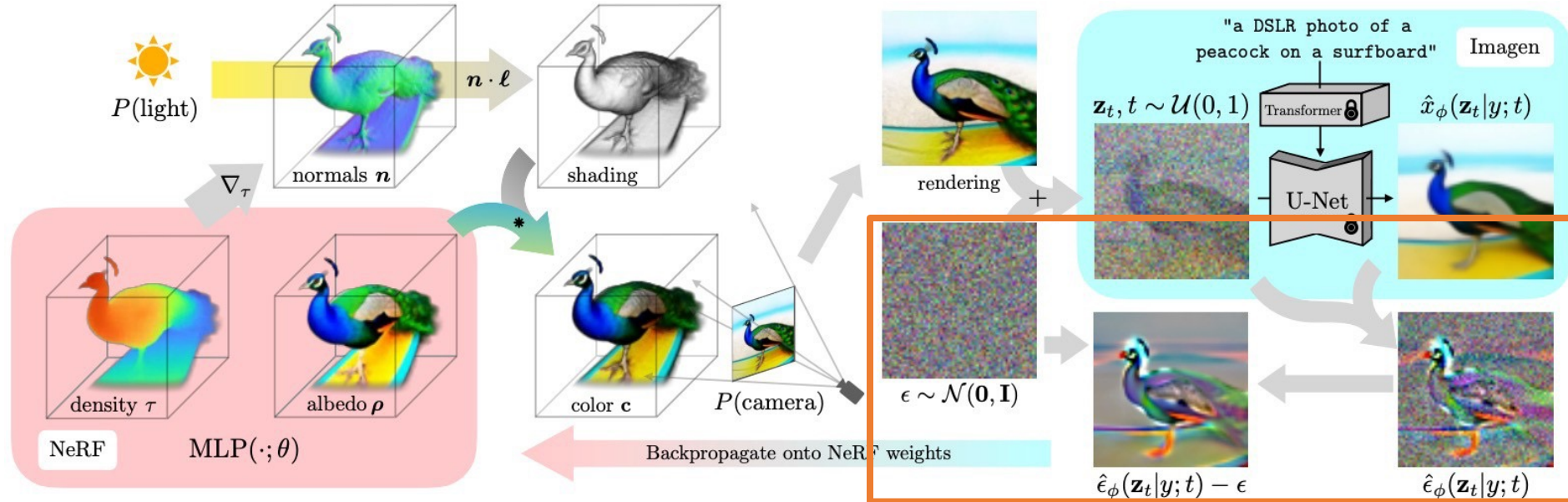
加上某個我們已知的 randomly sampled noise ϵ 去淹沒一些影像資訊

Score Distillation Sampling (SDS) 的直觀解釋



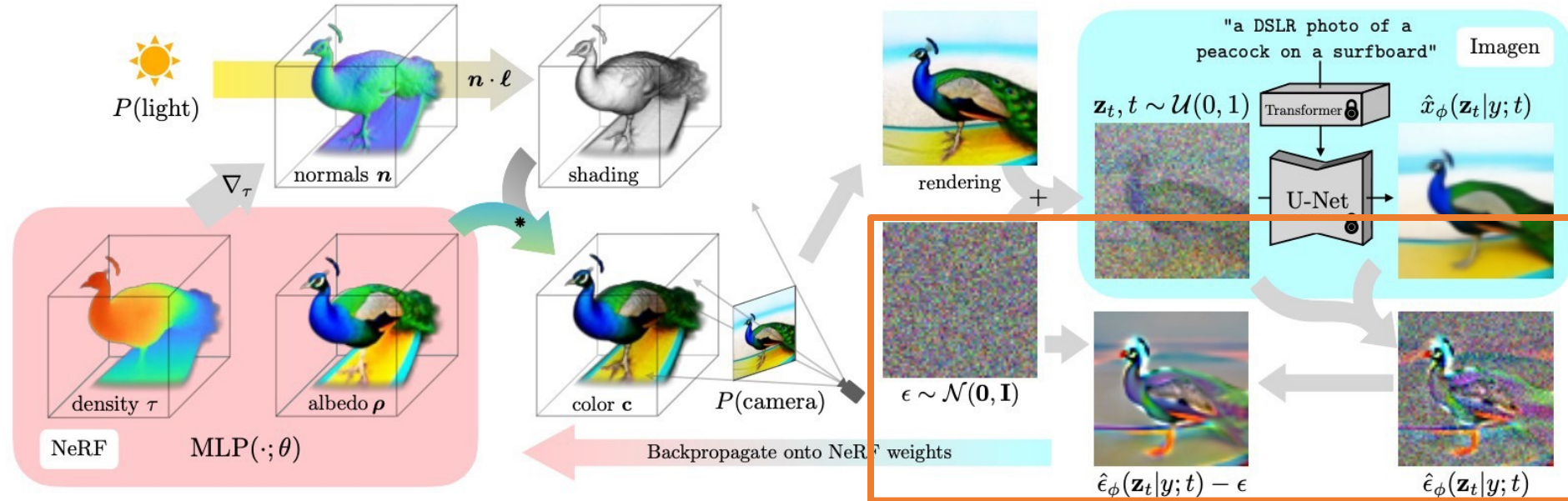
拿一個 pretrained diffusion model，他看過非常多真實影像
讓 diffusion model 去 predict noise (denoise)

Score Distillation Sampling (SDS) 的直觀解釋



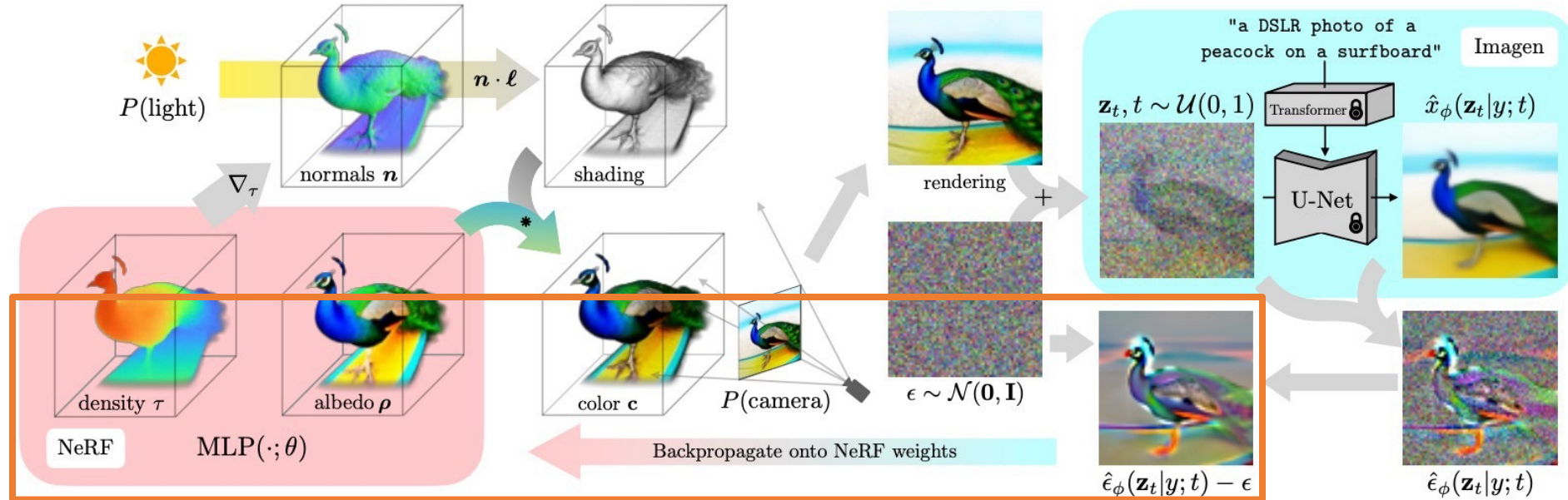
如果 predict 的 noise 跟一開始我們加上去的那個 noise ϵ 非常像
代表這張照片很接近真實影像的分佈
所以 pretrained diffusion 知道該怎麼 denoise 他

Score Distillation Sampling (SDS) 的直觀解釋



反之，如果 predict 的 noise 跟我們加上去的 noise 差很多
就代表這張照片根本不像一張真實照片
所以 diffusion model 不知道該怎麼 denoise 他

Score Distillation Sampling (SDS) 的直觀解釋



接著把 noise 的差異作為 loss 去更新 NeRF 的參數 θ

Connection Between DDIM Reverse Process and SDS

Recall: DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$
2. Compute $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t$.

Recall: DDIM

If $\sigma_t = \sqrt{1 - \bar{\alpha}_{t-1}}$ for all t (maximum variance),

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right)$
2. Compute $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \underset{0}{\sigma_t^2} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t).$
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.

Recall: DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$
2. Compute $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t$.

DDIM with Maximum Variance

If $\sigma_t = \sqrt{1 - \bar{\alpha}_{t-1}}$ for all t (maximum variance),

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right)$
2. Compute $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \underset{0}{\sigma_t^2} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t).$
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.

DDIM with Maximum Variance

If $\sigma_t = \sqrt{1 - \bar{\alpha}_{t-1}}$ for all t (maximum variance),

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right)$
2. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
3. Compute $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.

Connection to SDS

Use θ as the proxy variable encoding the information of $\mathbf{x}_{0|t}$.

For each time step $t = T, \dots, 1$, repeat: 在反向去噪過程中，time step t 是慢慢減小的

1. $\mathbf{x}_{0|t} = g(\theta; c).$

// (3) Compute $\mathbf{x}_{0|t}$ from θ

2. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$

例如：用 NeRF (參數 θ)
render 出一張照片 ($\mathbf{x}_{0|t}$)

3. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}}\mathbf{z}_t.$

4. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_{t-1}, t) \right).$

// (1) Compute $\mathbf{x}_{0|t-1}$

5. $\theta \leftarrow \theta - \frac{\partial}{\partial \theta} \|\mathbf{x}_{0|t-1} - g(\theta; c)\|^2.$

// (2) Update θ

Connection to SDS

Randomly sample the time step t (instead of decreasing).

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.
2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.
5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) \right)$.
6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

This becomes identical to SDS!

Why?

Q. 但 loss 看起來不太一樣？為什麼？

Connection to SDS

$$\begin{aligned} & \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2 \\ &= \left\| \frac{\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_{t-1}}} - \frac{\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t}{\sqrt{\bar{\alpha}_{t-1}}} \right\|^2 \\ &= \frac{1 - \bar{\alpha}_{t-1}}{\bar{\alpha}_{t-1}} \|\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) - \mathbf{z}_t\|^2 \end{aligned}$$

A. The same loss function, up to a scale!

Connection to SDS

Randomly sample the time step t (instead of decreasing).

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.
2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.
5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) \right)$.
6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

This becomes identical to SDS!

Why?

DDIM 反向去噪過程（生成）與 SDS 的差異：

1. SDS 的 time step t 是 randomly sample 的
2. SDS: Maximum Variance
3. DDIM 是 update $\mathbf{x}_{0|t}$ ，SDS 是 update 一個 proxy representation (例如 NeRF) 的參數 $\boldsymbol{\theta}$

Connection to SDS

SDS algorithm

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.
2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.
- ~~5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) \right)$.~~
6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) - \mathbf{z}_t\|^2$.

Connections Between DDIM and SDS

DDIM Reserve Process

SDS

Until convergence, repeat:

- For each time step $t = T, \dots, 1$, repeat: 
1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\mathbf{e}}_{\theta}(\mathbf{x}_t, t) \right)$
 2. Compute $\tilde{\mu} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \hat{\mathbf{e}}_{\theta}(\mathbf{x}_t, t)$.
 3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
 4. Compute $\mathbf{x}_{t-1} = \tilde{\mu} + \sigma_t \mathbf{z}_t$.
- SDS: Maximum Variance
1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.
 2. $\mathbf{x}_{0|t} = g(\theta; c)$.
 3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
 4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.
 5. $\theta \leftarrow \theta - \frac{\partial}{\partial \theta} \|\hat{\mathbf{e}}_{\theta}(\mathbf{x}_{t-1}, t) - \mathbf{z}_t\|^2$.
- DDIM 是 update $\mathbf{x}_{0|t}$, SDS 是 update 一个 proxy representation (例如 NeRF) 的参数 θ
- 

Connection to SDS

Randomly sample the time step t (instead of decreasing).

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.
2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \mathbf{z}_t$.
5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) \right)$.
6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

Q. 既然 DDIM 與 SDS 這麼像，我們能不能在使用 SDS 的時候模仿 DDIM denoising process 來讓結果更好？

例如：與其 randomly sample time step t ，為什麼不像 DDIM denoising process 一樣遞減？

Score Distillation via Inversion (SDI)

Can we change the variance σ_t instead of using the maximum σ_t ?

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.
2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) + \sigma_t \mathbf{z}_t$.
5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} (\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t))$.
6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

Q. 既然 DDIM 與 SDS 這麼像，我們能不能在使用 SDS 的時候模仿 DDIM denoising process 來讓結果更好？

或不要使用 maximum variance？因為 deterministic DDIM 的生成品質比較好

Score Distillation via Inversion (SDI)

What if we use $\sigma_t = 0$ (deterministic sampling)?

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.

2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.

~~3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.~~

4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1} - \cancel{\sigma_t^2}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) + \cancel{\sigma_t} \mathbf{z}_t$.

5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) \right)$.

6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

Score Distillation via Inversion (SDI)

We don't know $\mathbf{x}_t$ since the viewpoint c is randomly sampled.

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.

2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.

3. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$.

4. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}} \left(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t) \right)$.

5. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

我們想要 deterministic sampling ($\sigma_t = 0$)
但我們不知道 $\mathbf{x}_t$ ，因為視角 c 一直在變

用同一個 initial noise $\mathbf{x}_t$ 來表示不同視角的圖片
並不合理，該怎麼辦？

Score Distillation via Inversion (SDI)

Solution: DDIM Inversion

Until convergence, repeat:

1. $t \sim \mathcal{U}(1, T)$. $c \sim \mathcal{U}(\mathcal{C})$.

2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.

3. $\mathbf{x}_t = \text{DDIM_Inversion}(\mathbf{x}_{0|t}, t)$.

A.

既然每個視角 c 的 $\mathbf{x}_t$ 都不一樣，

那就每個 iteration 都用 DDIM inversion 從 $\mathbf{x}_{0|t}$ 推估出 $\mathbf{x}_t$

4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$.

5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}}(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t))$.

6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

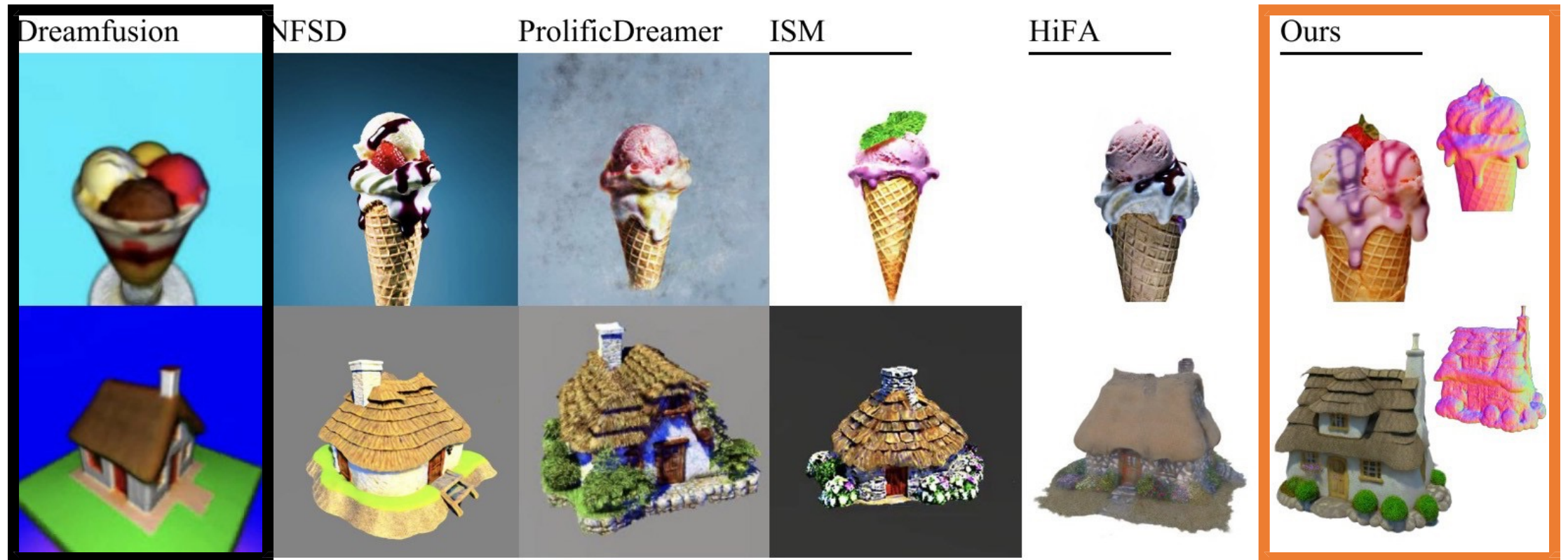
Score Distillation via Inversion (SDI)

Additionally, decrease the time step t like the reverse process.

For each time step $t = T, \dots, 1$, repeat:

1. $c \sim \mathcal{U}(\mathcal{C})$.
2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
3. $\mathbf{x}_t = \text{DDIM_Inversion}(\mathbf{x}_{0|t}, t)$.
4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$.
5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}}(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t))$.
6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

Score Distillation via Inversion (SDI)



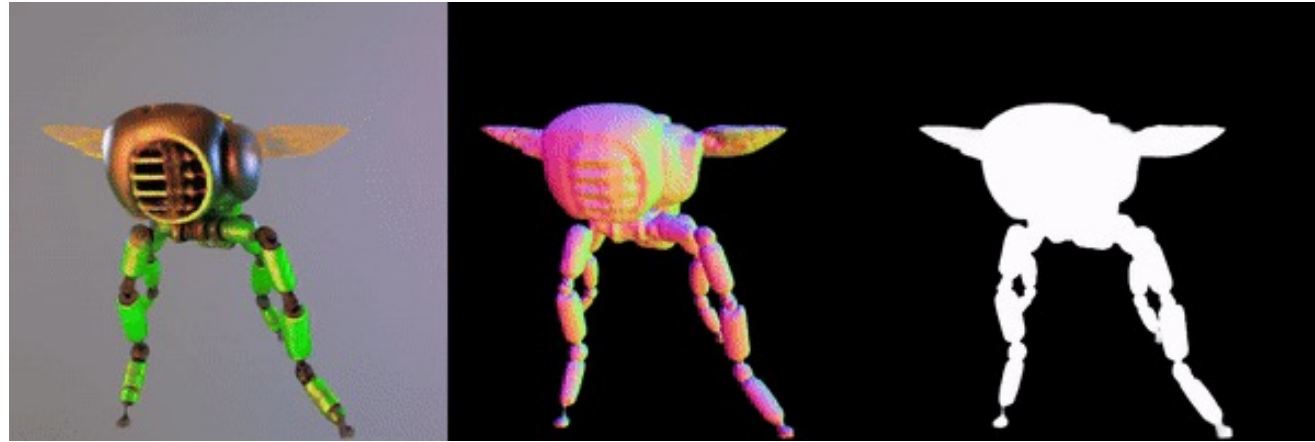
SDS

SDI

優點：生成品質變好，因為更貼近 deterministic DDIM 的反向去噪過程

缺點：生成時間變長，因為每個 iteration 都要做 DDIM inversion

Score Distillation via Inversion (SDI)



“Robotic bee high detail”

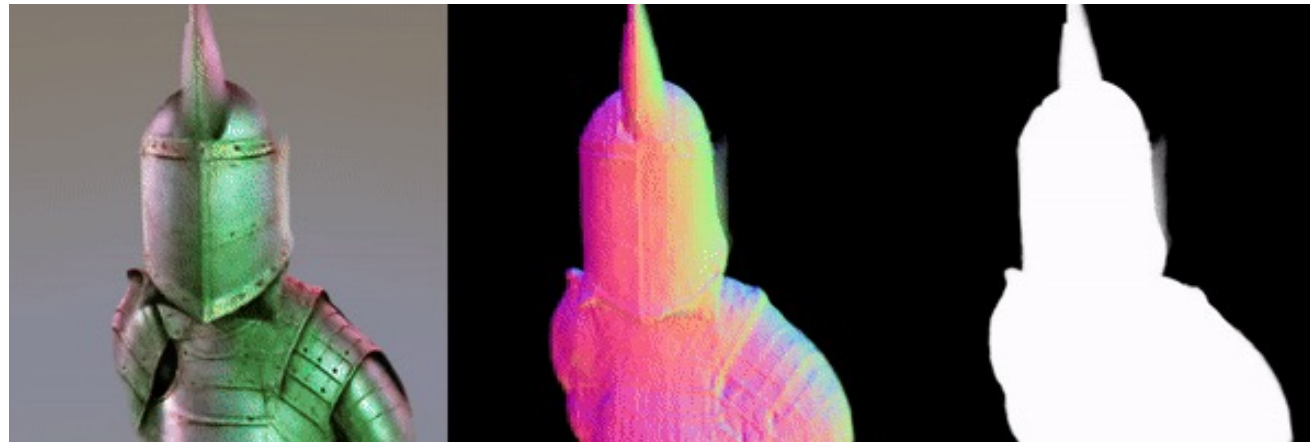


“Pumpkin head zombie, skinny, highly detailed”

Score Distillation via Inversion (SDI)



“An ice cream sundae”



“A photograph of a knight”

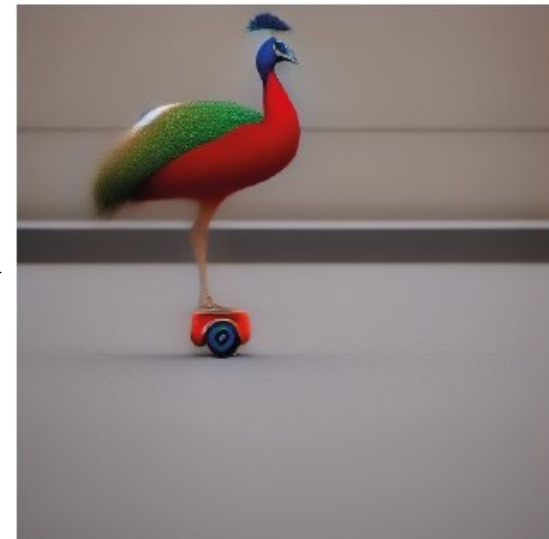
Extension to Editing

Limitations of SDS

In **editing**, SDS tends to lose the identity of the input image.



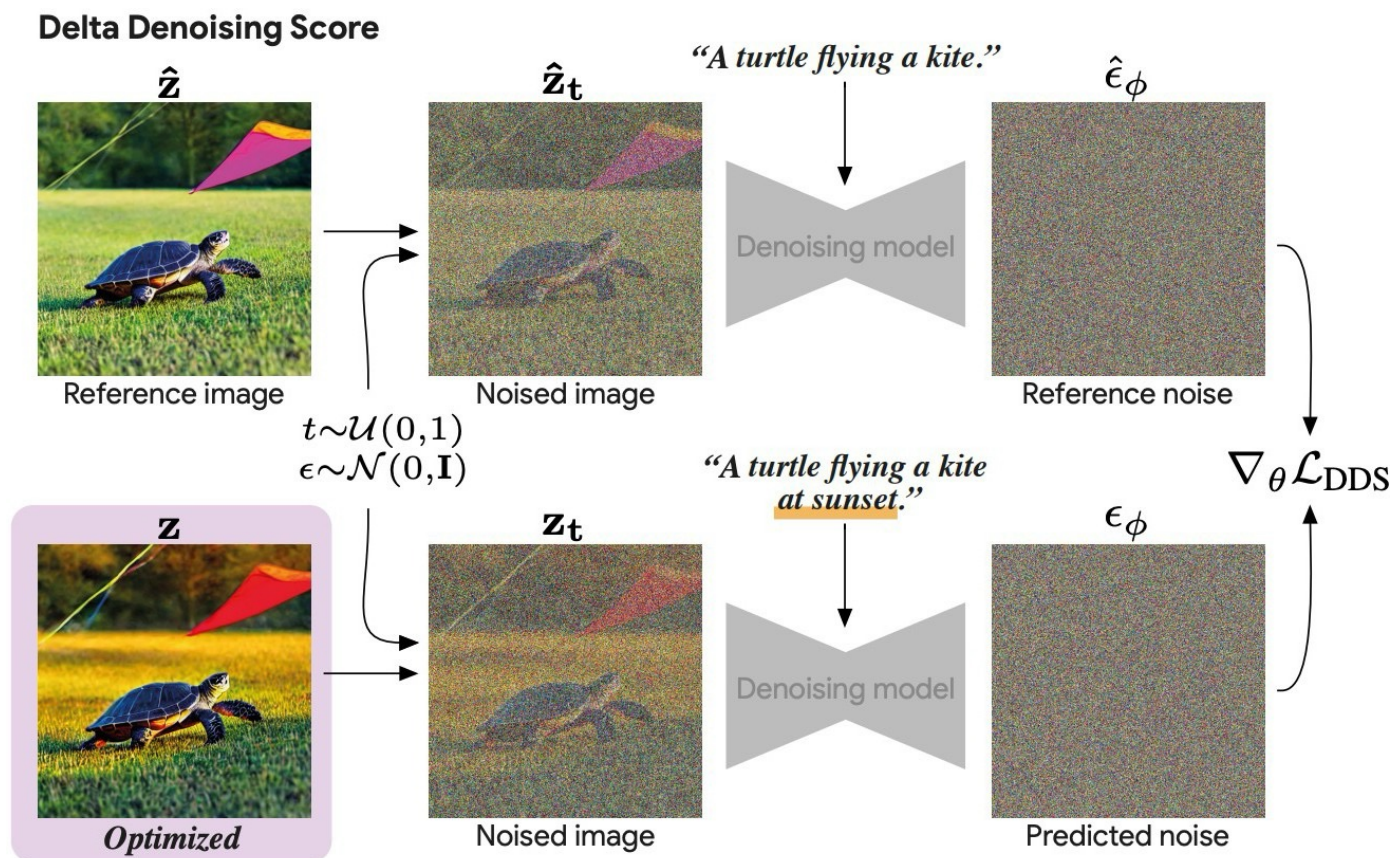
A flamingo is
roller-skating in the city.



Editing 過的影像全變了，幾乎跟全新生成的一樣，沒什麼輸入影像的資訊
Q. 該怎麼保留輸入影像的資訊？

Delta Denoising Score

A modification of **SDS** for text-driven **editing**, not generation.

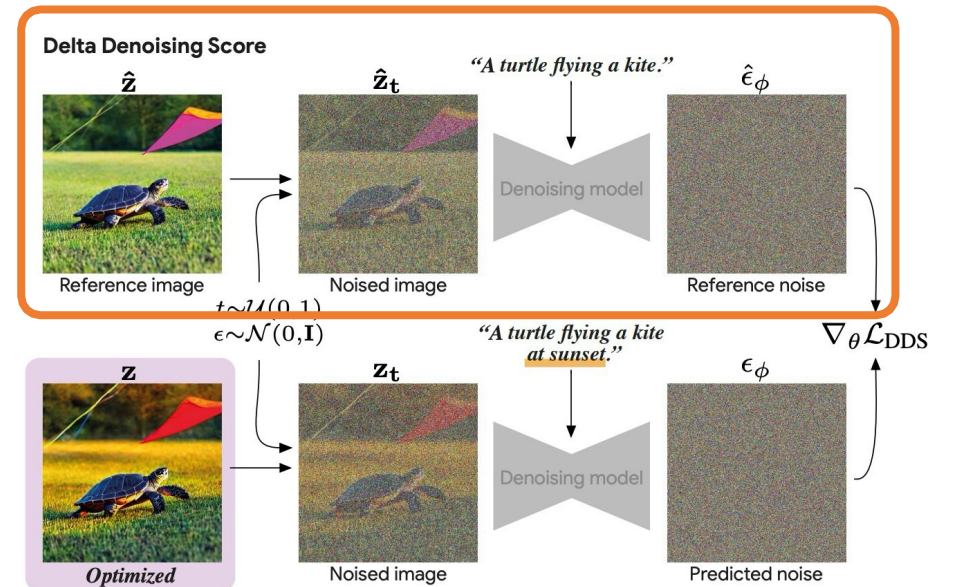


Delta Denoising Score

Given a **reference** image $\mathbf{x}_0^{\text{ref}}$, **edits** the reference image by **matching** the predicted noise (score) of the updated and reference images:

$$\mathcal{L}_{\text{DDS}}(\mathbf{x}_0; \mathbf{x}_0^{\text{ref}}) = \mathbb{E}_{t, \epsilon} \left[\left\| \epsilon_{\phi}(\mathbf{x}_t, y, t) - \epsilon_{\phi}(\mathbf{x}_t^{\text{ref}}, y^{\text{ref}}, t) \right\|_2^2 \right]$$

- $\mathbf{x}_0^{\text{ref}}$: Reference image.
- y^{ref} : Text prompt of the reference image.
- $\mathbf{x}_0$: Updated image at each step.
- y : New text prompt for editing.



Delta Denoising Score

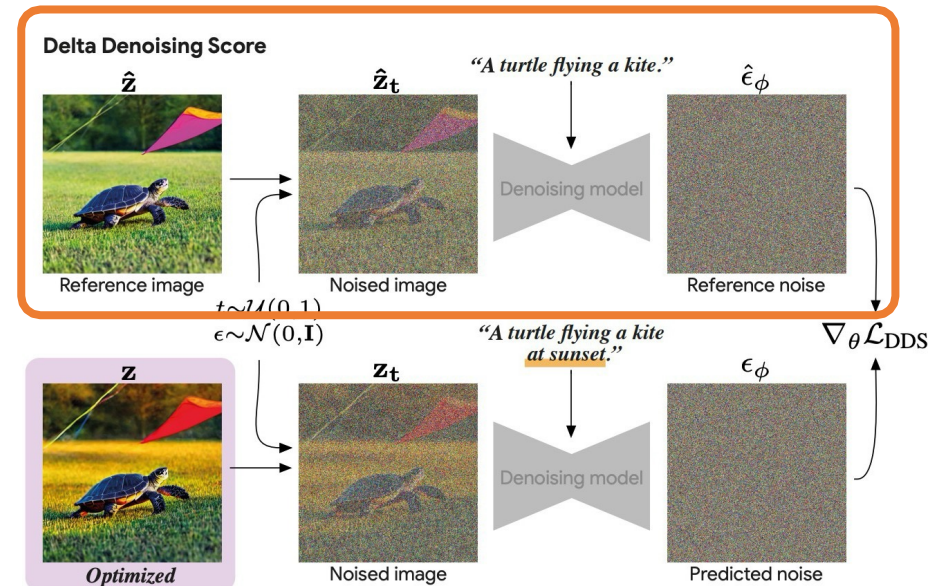
Given a **reference** image $\mathbf{x}_0^{\text{ref}}$, **edits** the reference image by **matching** the predicted noise (score) of the updated and reference images:

$$\mathcal{L}_{\text{DDS}}(\mathbf{x}_0; \mathbf{x}_0^{\text{ref}}) = \mathbb{E}_{t, \epsilon} \left[\left\| \epsilon_{\phi}(\mathbf{x}_t, y, t) - \epsilon_{\phi}(\mathbf{x}_t^{\text{ref}}, y^{\text{ref}}, t) \right\|_2^2 \right]$$

Use the same noise ϵ .

- $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$

- $\mathbf{x}_t^{\text{ref}} = \sqrt{\bar{\alpha}_t} \mathbf{x}_0^{\text{ref}} + \sqrt{1 - \bar{\alpha}_t} \epsilon$



Delta Denoising Score

DDS also often fails to preserve **identity**.

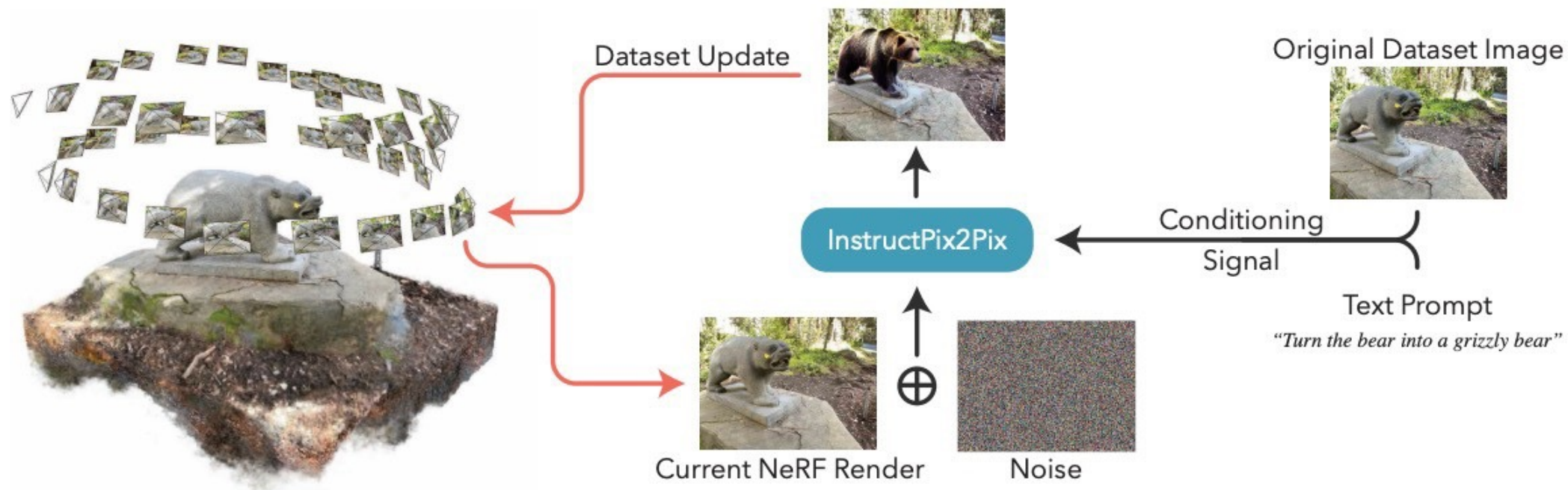
Delta Denoising Score

DDS also often fails to preserve identity and sometimes fail **entirely**.

Instruct-NeRF2NeRF

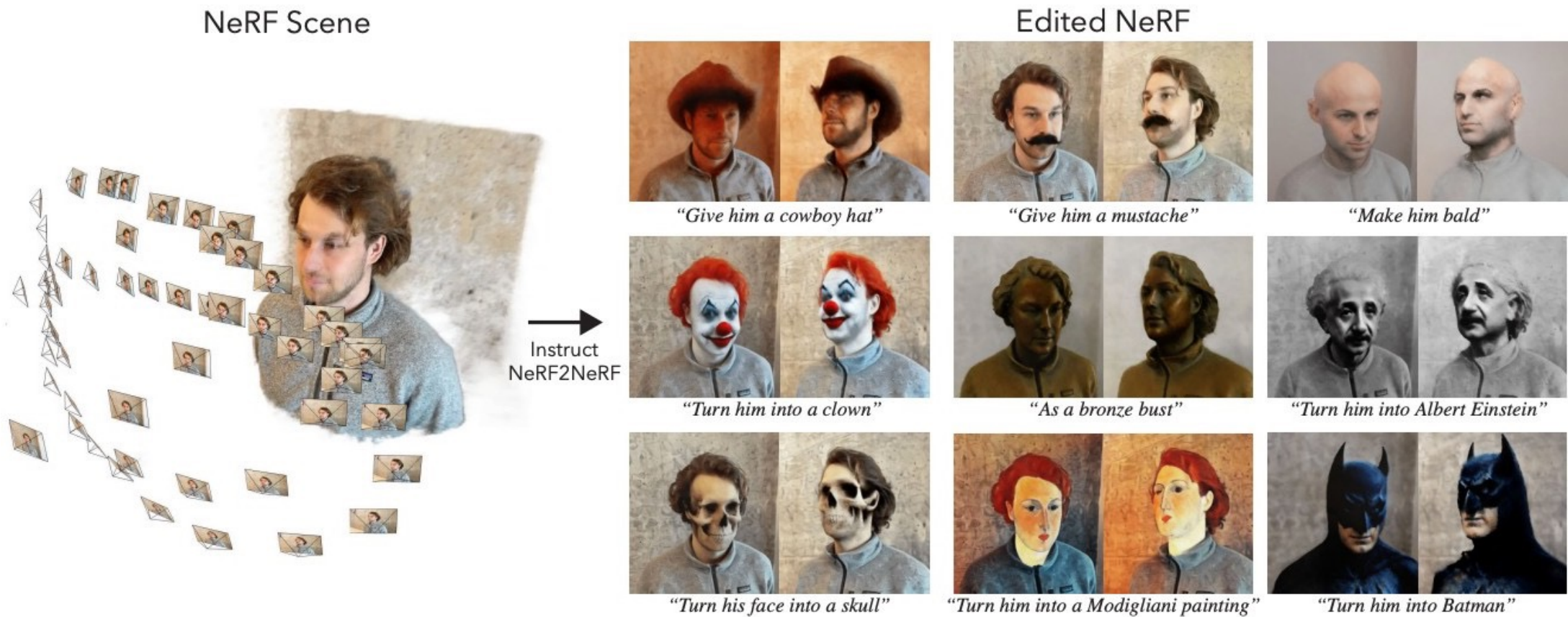
Iterative Data Update (IterativeDU):

Iteratively **replace** each multi-view image using InstructP2P and **update** the NeRF reconstruction.



Instruct-NeRF2NeRF

Impressive results in appearance changes.

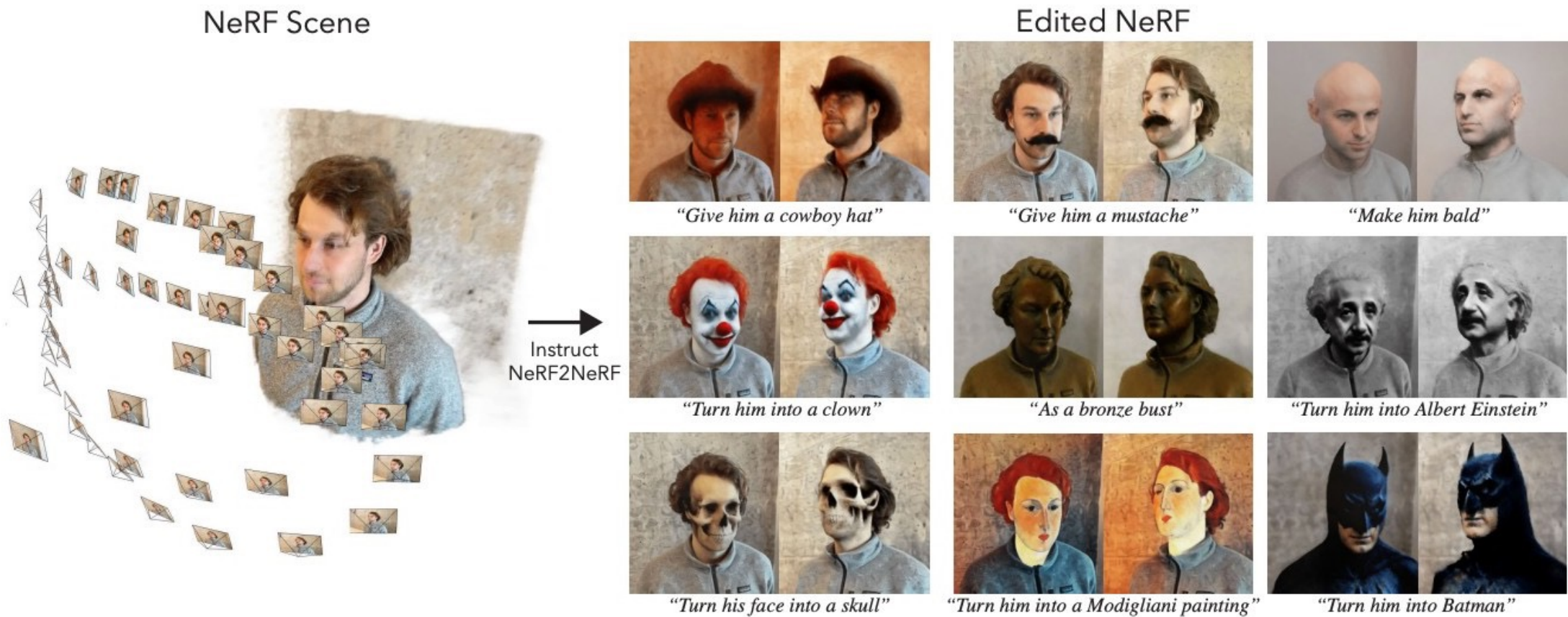


Instruct-NeRF2NeRF – Limitations

- IterativeDU methods cannot induce **geometric** changes.
- E.g., changing the pose of a human body.

Instruct-NeRF2NeRF

Impressive results in appearance changes.

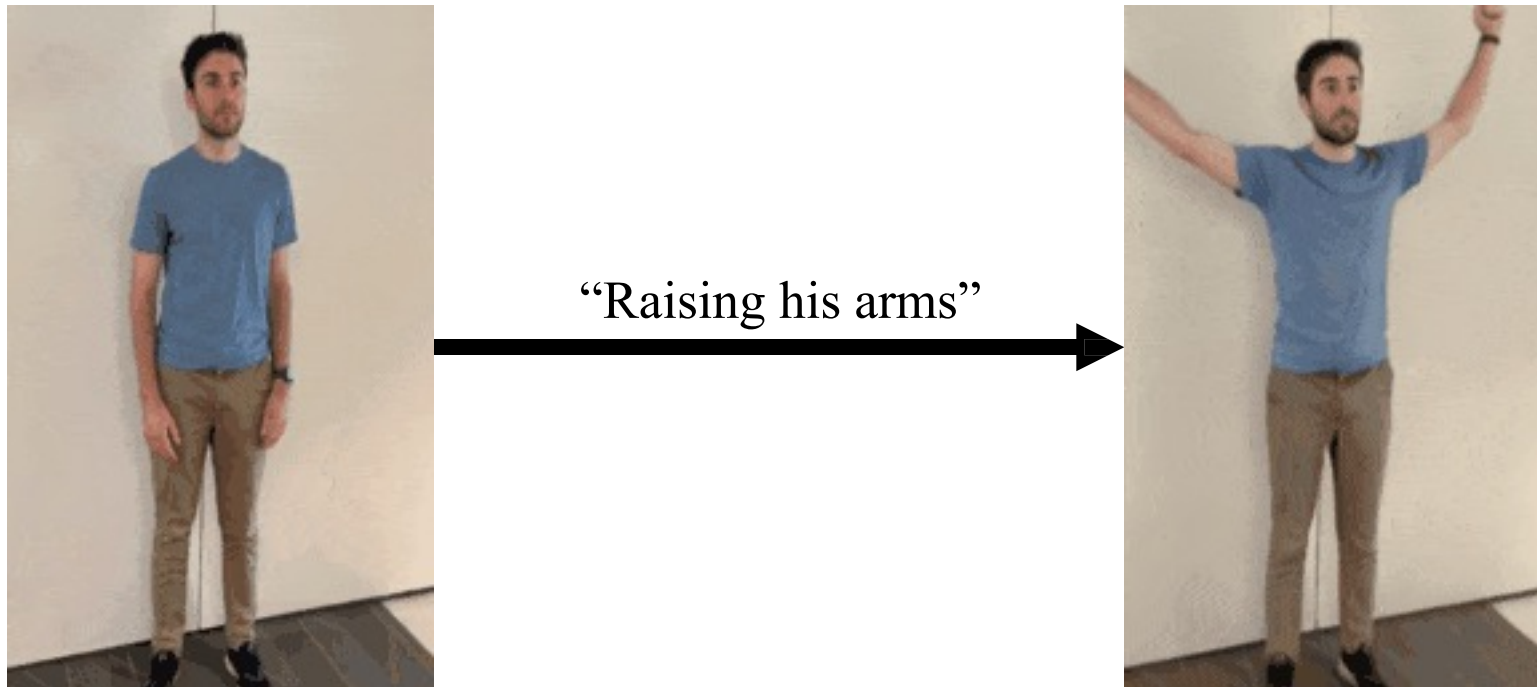


Instruct-NeRF2NeRF – Limitations

- IterativeDU methods cannot induce **geometric** changes.
- E.g., changing the pose of a human body.

Posterior Distillation Sampling (PDS)

A modification of SDS for text-driven NeRF editing enables proper **geometric** and **appearance** changes.



Posterior Distillation Sampling (PDS)

A modification of SDS for text-driven NeRF editing enables proper **geometric** and **appearance** changes.



“With hot air balloons”



Posterior Distillation Sampling (PDS)

A modification of SDS for text-driven NeRF editing enables proper **geometric** and **appearance** changes.



“Leonardo DiCaprio”

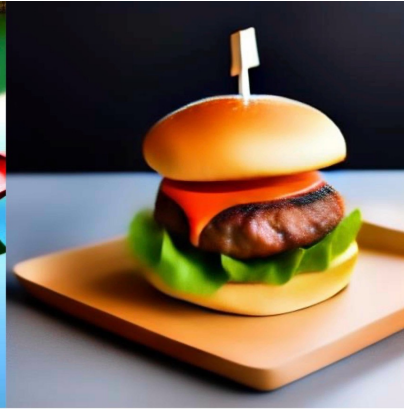


Assignment 3

SDS, VSD, and SDI Implementations



"A boat on a river"



"A burger on the table"



"A car on the road"



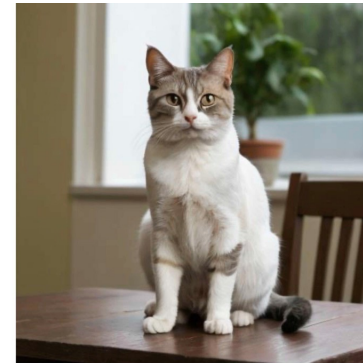
"A cat sitting on the table"



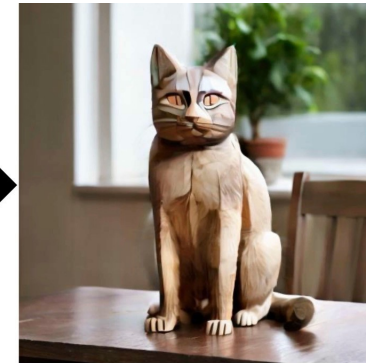
"A castle next to a river"



"A rainbow castle next to a river"



"A cat sitting on a table"



"A wooden cat statue sitting on a table"